

INSTANT LESSONS



for senior maths

Sue Thomson
Ian Forster
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CLASSMASTERS

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Preface

This is a collection of reproducible worksheets created to address the changing emphasis in modern senior Mathematics courses. Students will meet questions requiring good graphical skills and an understanding of concepts, and will need to apply theory to realistic situations.

Several different styles of activities have been incorporated in this book, including:

- worksheets for individual students
- classroom activities for group work or individual study
- creative teaching ideas to promote interest and understanding

We have included activities designed to lead the students through areas that traditionally cause them difficulties. On their own, in a graded exercise, such tasks may cause no difficulties, but students need to meet questions in situations that require them to identify the types of problems, and then solve them. To address this, some activities involve a mixture of questions.

The worksheets are grouped in topic areas to make them easy to use. They are arranged in the normal teaching sequence within each topic. The skills developed in each activity, how the activity can be used, the appropriate student level and the expected time for completion are all given in the contents.

These *Instant Lessons* will bring variety to the classroom, and will provide students with worksheets for consolidation and revision. Also they can quickly and easily be prepared to provide work for a class when the regular teacher is absent.

Sue Thomson, Ian Forster and Narelle Nelson

Contents

The study sheets in this book have been designed to help students with a systematic revision of Year 12 Mathematics. In the following list, the title of each activity or study sheet is accompanied by a description of its contents. Codes indicate the abilities of the students for whom the activities are suited and include the number of minutes required to complete each activity. The codes are:

A: Low average **B:** Average **C:** Capable

For example, code B45 indicates that the activity is pitched at an average level and typical students should require 45 minutes to complete it. Naturally the suggested time should be taken only as a guide.

Instructions for the teacher: Each cardmatch and squaresaw activity should be photocopied onto a different coloured card and then cut out. Students can work in small groups, with one set of cards per group.

FACTORISATION AND EXPANSION

CODE

5-1	Factorisation classification — cardmatch (2 pages)	A card activity to help students recognise the different methods of factorising. Can also be used to revise factorisation.	A/B45
5-2	Factorisation/expansion squaresaw	Revises factorisation and/or expansion.	A/B20

LOGARITHMS

5-3	Expansions of log statements — cardmatch	A card activity to help revise log laws.	B15
5-4	Log squaresaw	Revises log and index laws.	B15

PARALLEL AND PERPENDICULAR LINES

5-5	Parallel and perpendicular lines classification	A card activity to practise finding gradients of lines and testing for parallel and perpendicular lines.	A/B20
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PARABOLAS

5-6	Features of the parabola — cardmatch	A card activity on parabolas involving vertex, focal length, focus, directrix and graph.	A/B30
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GRAPHS AND EQUATIONS

5-7	Graph classification — cardmatch (2 pages)	Matching equations and standard graphs with their appropriate graph types.	A/B45
5-8	Matching graphs — cardmatch (2 pages)	Matching equations with their graphs.	B/C45

DERIVATIVES

5-9	Derivatives — cardmatch	Matching log, exponential and trig functions with their derivatives.	B30
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USING THE DISCRIMINANT

CODE

5-10	Discriminant — cardmatch (2 pages)	Matching the discriminant, the number of roots and the graph for 6 different quadratic equations.	A/B30
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MOTION

5-11	Investigating motion (2 pages)	Investigates displacement, velocity and acceleration in displacement–time graphs, velocity–time graphs and acceleration–time graphs.	B40
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INTEGRATION AND DIFFERENTIATION

5-12	Integration/differentiation — cardmatch	A cardmatch activity involving differentiating and integrating trig functions.	B30
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GEOMETRIC SERIES

5-13	A series of questions	Four different types of questions requiring a knowledge of geometric series.	C40
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PARABOLA LANGUAGE

5-14	Parabola language — cardmatch (2 pages)	Matching diagrams with word descriptions involving the parametric parabola $x = 2ap$, $y = ap^2$.	B20
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REVISION

5-15	Study sheet 1	} Miscellaneous study sheets that are useful for revision before exams or as worksheets, especially if a teacher is absent from school.	B45
5-16	Study sheet 2		B/C45
5-17	Study sheet 3		B/C60
5-18	Study sheet 4		B/C60
5-19	Study sheet 5		B60
5-20	Study sheet 6		B60
5-21	Study sheet 7		B/C60
5-22	Study sheet 8		B/C60
5-23	Study sheet 9		B/C60
5-24	Study sheet 10		B/C60
5-25	Study sheet 11	A parabola revision sheet.	B60
5-26	Study sheet 12 (3 pages)	A holiday assignment.	B90
5-27	Study sheet 13 (2 pages)	Mixed revision.	B90
5-28	Study sheet 14	More mixed revision.	B60

ANSWERS

pages 43–48

Factorisation classification – cardmatch

5-1A

Students can work in small groups with one set of cards per group.

Students are required to match *each expression* with the *type of factorisation* required to factorise it. A follow-up activity could be to actually factorise each expression.

1

Common factor

2

Grouping in pairs

3

Difference of two squares

4

Combination

5

Trinomials

6

Cubic

A

$$8x^3 - 1$$

B

$$cx - cy + fx - fy$$

C

$$80m^2 - 20n^2$$

D

$$x^2 - 6x$$

E

$$x^2 + xy + 7x + 7y$$

F

$$5xy + 15yz$$

G

$$m^2 - n^2 + 5m - 5n$$

H

$$10a + 25b$$

I

$$12y - 6y^2$$

J

$$1 - 81a^2$$

K

$$a^2 - 11a + 24$$

L

$$b^3 - 1$$

Factorisation classification – cardmatch (continued)

5-1B

M $3ax - 2bx + 3ay - 2by$	N $9x^2 - 49y^2$	O $a^3 + 125$
P $3x^2 + 12xy$	Q $x^2 - 5x - 14$	R $21a^2 + 7ab - 35a$
S $27 - y^3$	T $x^3 - 5x^2 + 4x - 20$	U $x(a - b) + 3(b - a)$
V $ay^2 + 4ay - 3y$	W $42a^2 - 17a - 15$	X $(x - 3y)^2 - 16y^2$
Y $12x - 6y + 2mx - my$	Z $x^3 - 9x - x^2 + 9$	AA $14x^2 - 80x - 24$
BB $6a^3 + 6$	CC $(x - 3)^2 - 16$	DD $9y^3 + 72$
EE $1 - x - 72x^2$	FF $x^6 - 1$	GG $a^2b^2 - 1$
HH $10x^2 - x - 3$	II $x^4 - 81$	JJ $10a^2 + 72a + 14$

Factorisation/expansion squaresaw

5-2

Students are required to match the factors and the expansions, just like completing a jigsaw.

$3a(a^2 + 1)$ B $2a^3 - 8a$ $(a - 6)(a - 1)$	$2a(a - 2)(a + 2)$ J $a^2 - 9$ $a^2 + a$	$(a - 3)(a + 3)$ D $a^3 - a$ $(a - 1)(a^2 + a + 1)$	$a(a - 1)(a + 1)$ G $a^3 + 6a$ $a^2 + 2a + 1$
$a^2 - 7a + 6$ N $5(3 + 2a)$ $4 - 16a^2$ $(x + 2)(a + b)$	$a(a + 1)$ E $4(1 - 2a)(1 + 2a)$ $3a^2 + 8a - 16$ $(a - 2)(a^2 + 2a + 4)$	$a^3 - 1$ P $a^3 - 8$ $a^2 - 2a + 1$ $a^2 - ab - ac + bc$	$(a + 1)^2$ R $5a^2 + 6a$ $(3a - 2)(a - 1)$ $(a - b)(a - c)$
$a(x + 2) + b(x + 2)$ A $(a - 7)(a + 7)$ $4a^2 - 4a - 15$ $4a^2 - 16a - 9$	$(3a - 4)(a + 4)$ F $(2a + 1)(2a - 9)$ $2a + 2b$ $4a^2 - 25$	$(a - 1)^2$ L $(2a - 5)(2a + 5)$ $5a^3 + 10a^2b$ $a^2 + 7a + 6$	$3a^2 - 5a + 2$ K $a^2 - a - 6$ $a^3b - ab^3$ $(a + 6)(a + 1)$
$(2a - 5)(2a + 3)$ H $ab(a - b)$ $a^2 - b^2$	$2(a + b)$ C $(a - b)(a + b)$ $a^3 + 1$	$5a^2(a + 2b)$ M $(a + 1)(a^2 - a + 1)$ $8a^2b - 3a^2b^2$	$ab(a - b)(a + b)$ T $a^3 - 27$ $a^2b(8 - 3b)$

Expansions of log statements – cardmatch

5-3

Students are required to match *each expression* with its *expanded form*.

$\log_e 3xy$ C	$\log_e 3 + \log_e x + \log_e y$ 4
$\log_e x^3y$ A	$3 \log_e x + \log_e y$ 3
$\log_e \frac{3x}{y}$ G	$\log_e 3 + \log_e x - \log_e y$ 1
$\log_e xy^3$ B	$\log_e x + 3 \log_e y$ 8
$\log_e (xy)^3$ J	$3 (\log_e x + \log_e y)$ 6
$\log_e \sqrt[3]{xy}$ F	$\frac{1}{3} (\log_e x + \log_e y)$ 2
$\log_e \frac{3}{xy}$ D	$\log_e 3 - \log_e x - \log_e y$ 11
$\log_e 3xy^3$ I	$\log_e 3 + \log_e x + 3 \log_e y$ 5
$\log_e \sqrt[3]{x^3y}$ E	$\log_e x + \frac{1}{3} \log_e y$ 7
$\log_e \frac{x^3}{y}$ H	$3 \log_e x - \log_e y$ 10
$\log_e \sqrt{3xy}$ K	$\frac{1}{2} (\log_e 3 + \log_e x + \log_e y)$ 9

Log squaresaw

5-4

Students are required to match each log or index expression with its equivalent expression. They should not use calculators.

1	2 A $\log_e \sqrt{e}$	$\log_e 0$	undefined	J $\log_e e$	$\log_2 8$	3	M $\sqrt{e^6}$	$\log_{10} 2 + \log_{10} 5$	3 L $\log_e 14 - \log_e 2$	$\log_4 16$
	$\frac{1}{2}$ K $\log_e \frac{1}{x}$	$e^{\log_e 3}$	3	G $\log_e \frac{1}{e^2}$	$\frac{1}{\log_e \sqrt{e}}$	$\frac{1}{-2}$ B $\log_e 1$	e^3	$(\log_5 25)\sqrt{5}$	$\log_e 7$	$2\frac{1}{2}$ E $\log_e x^3$
	$-\log_e x$ C $\log_3 1 - \log_3 9$	$\log_e \sqrt{x}$	$\frac{1}{2} \log_e x$	H $\log_e e^2$	$\log_5 25$	2 R $\log_e 3 + \log_e 4$	0	3 $\log_{10} 8 + \log_{10} 125$	$3 \log_e x$	D $\frac{\log_e 8}{\log_e 2}$
2	-2 P $\log_e \frac{1}{e}$	-1	2	N $\log_e 5e$	$1 + \log_e 5$	$\log_e 12$ F $\log_5 5\sqrt{5}$	3	Q		

Parallel and perpendicular lines classification

5-5

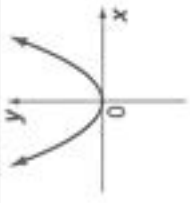
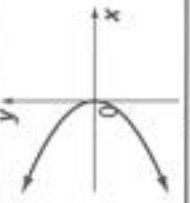
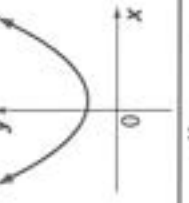
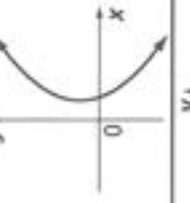
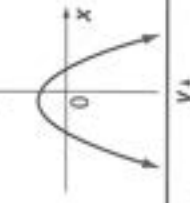
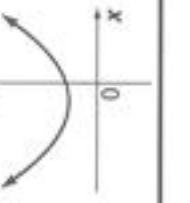
In this cardmatch, students are required to match *each pair of lines* with its *appropriate heading*, i.e. 'Parallel lines' or 'Perpendicular lines'.

Parallel lines	Perpendicular lines
$2x - y + 4 = 0$ and $6x - 3y + 7 = 0$ A	$x + 3y - 2 = 0$ and $3x - y + 4 = 0$ H
$y = 5x + 2$ and $y = 5x - 7$ F	$7x + 3y - 8 = 0$ and $3x - 7y + 1 = 0$ D
$3x + 2y = 7$ and $3x + 2y - 16 = 0$ G	$x + 2y = 5$ and $y = 2x + 3$ B
$y + 2x = 3$ and $y = -2x + 5$ I	$3x - 4y + 5 = 0$ and $4x + 3y - 8 = 0$ K
$y = -2x + 3$ and $y = 5 - 2x$ C	$y = 2x + 3$ and $x + 2y = 10$ J
$y + 4x - 5 = 0$ and $2y = -8x + 1$ L	$2y - 5x = 7$ and $5y = 1 - 2x$ E

Features of the parabola – cardmatch

5-6

In this cardmatch, students are required to match *the five features of a parabola* with the appropriate equation.

	Equation of parabola	Vertex	Focal length	Focus	Equation of directrix	Graph of parabola
1	$x^2 = 16y$	(0, 0)	$a = 4$	(0, 4)	$y = -4$	
2	$y^2 = -20x$	(0, 0)	$a = 5$	(-5, 0)	$x = 5$	
3	$(x - 1)^2 = 8(y - 3)$	(1, 3)	$a = 2$	(1, 5)	$y = 1$	
4	$(y - 2)^2 = 12(x - 2)$	(2, 2)	$a = 3$	(5, 2)	$x = -1$	
5	$x^2 + 4y + 2x = 11$	(-1, 3)	$a = 1$	(-1, 2)	$y = 4$	
6	$x^2 + 4x - 12y + 40 = 0$	(-2, 3)	$a = 3$	(-2, 6)	$y = 0$	

Graph classification – cardmatch

5-7A

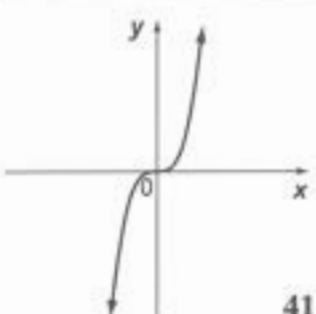
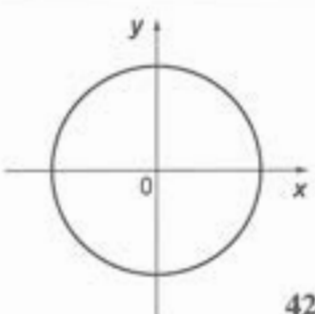
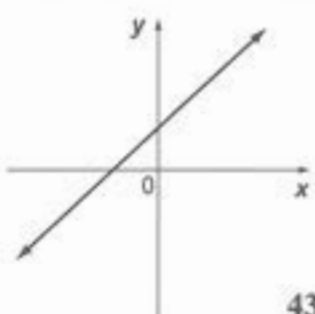
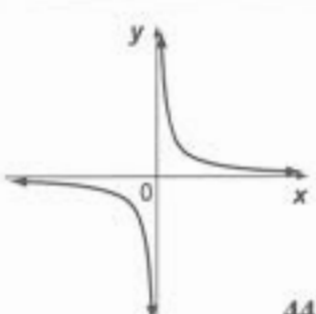
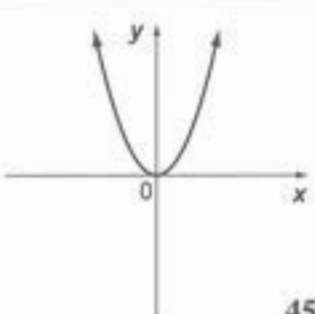
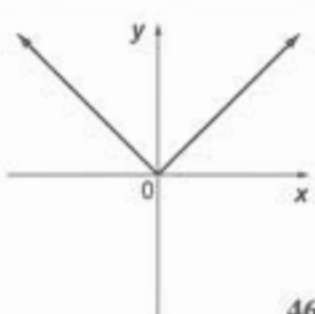
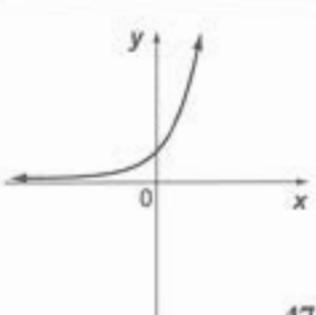
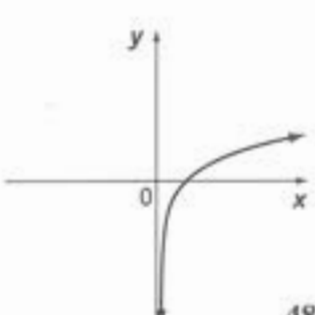
In this cardmatch activity, students are required to match the *equation of a curve* with its *name* and *standard graph*.

straight line	parabola	hyperbola	circle or semicircle
cubic	exponential	logarithmic	absolute value
$y = x + 3$ 1	$y = x^2$ 2	$y = \frac{2}{x}$ 3	$y = -\sqrt{16 - x^2}$ 4
$h(x) = x^3$ 5	$y = 2 - x$ 6	$f(x) = x^2 + 1$ 7	$y = \frac{1}{x}$ 8
$g(x) = 3 - x^2$ 9	$y = 2^x$ 10	$x^2 + y^2 = 16$ 11	$xy = 5$ 12
$y = x^3 + 2$ 13	$2x + 3y = -6$ 14	$y = 3x - 1$ 15	$y = 2^{-x}$ 16
$f(x) = x^2 + x$ 17	$y = \sqrt{4 - x^2}$ 18	$x = 2y + 1$ 19	$y = \frac{3}{x}$ 20
$y = \log_e(x - 1)$ 21	$g(x) = \sqrt{9 - x^2}$ 22	$x = 3$ 23	$y = -\frac{2}{x}$ 24
$y = x $ 25	$x^2 + y^2 = 1$ 26	$y = x$ 27	$x^2 + (y - 1)^2 = 4$ 28

Graph classification – cardmatch (continued)

5-7B

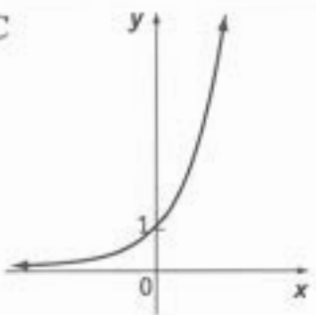
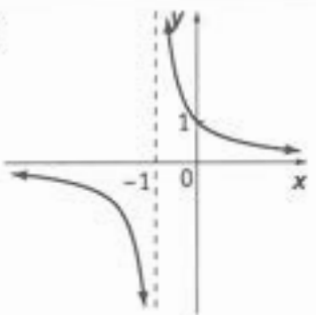
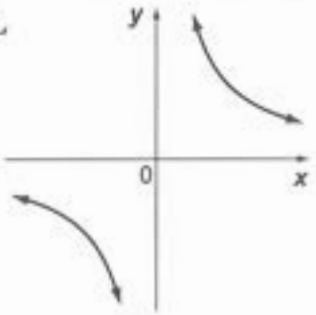
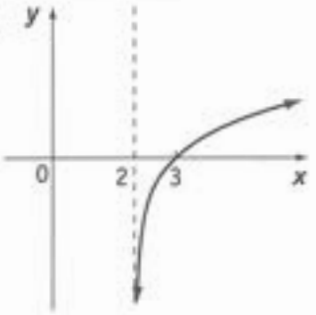
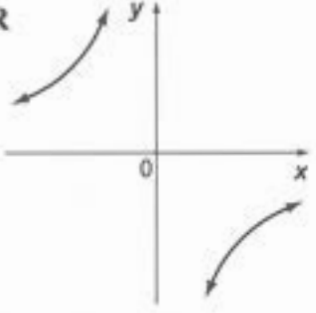
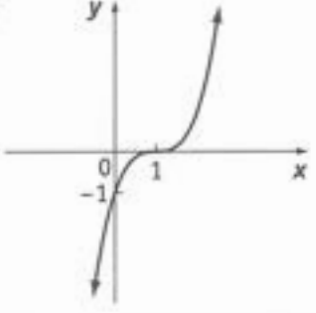
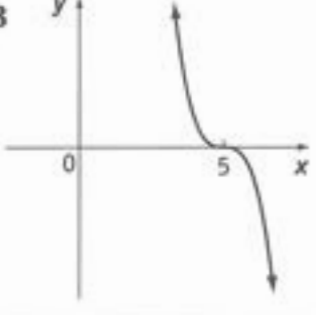
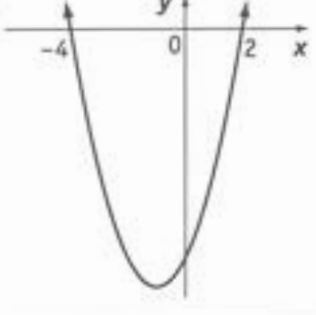
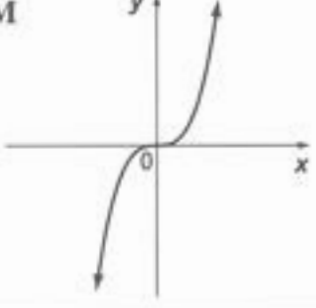
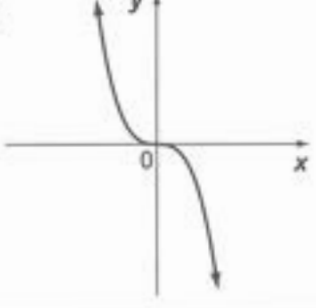
$xy = -3$ 29	$y = e^x$ 30	$y = \log_{10} x$ 31	$y = \frac{-1}{2x+1}$ 32
$y = x^2 - 3x + 1$ 33	$y = x^3 - 2$ 34	$y = 0$ 35	$y = 9 - x^2$ 36
$y = x + 1 $ 37	$y = \log_e 3x$ 38	$y = 9 - x^3$ 39	$y = 2x + 1 $ 40

 41	 42	 43
 44	 45	 46
 47	 48	

Matching graphs – cardmatch

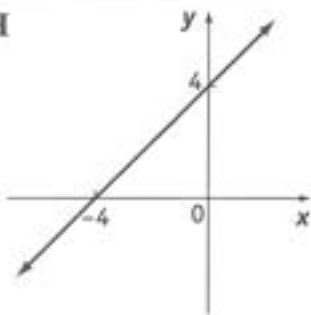
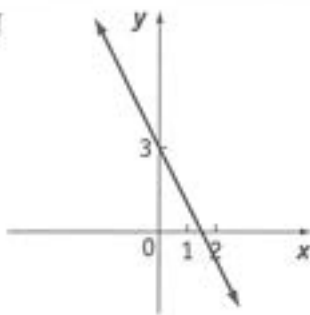
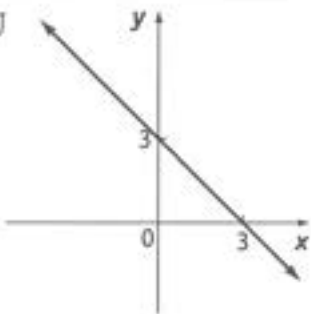
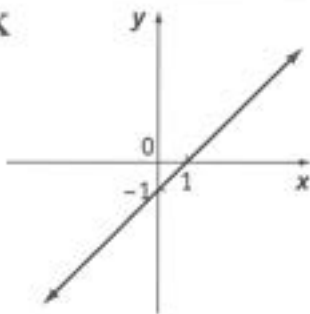
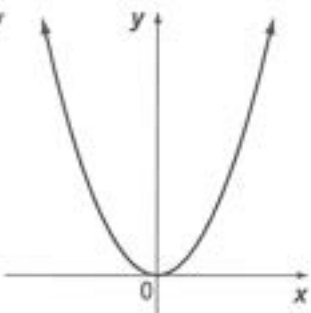
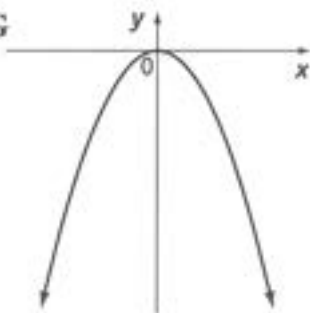
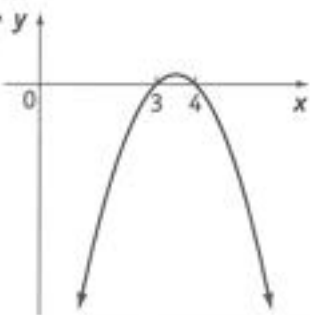
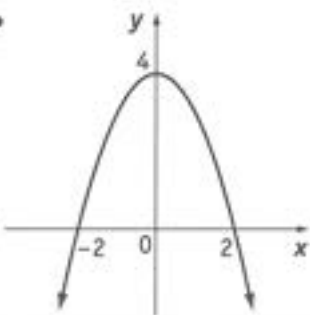
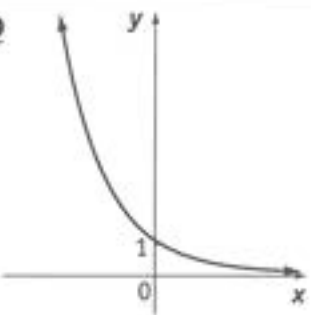
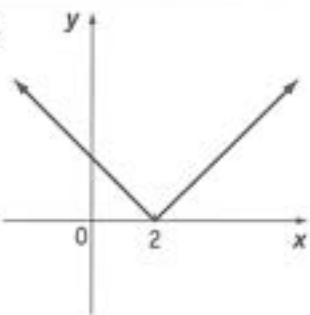
5-8A

In this cardmatch activity, students are required to match the *equation of a curve* with its *graph*.

C 	E 	1 $y = 3^x$
L 	A 	3 $y = \frac{4}{x}$
R 	F 	5 $y = \frac{-5}{x}$
B 	N 	7 $y = (5 - x)^3$
M 	D 	9 $y = x^3$
		10 $y = -x^3$
		4 $y = \log_e(x - 2)$
		6 $y = (x - 1)^3$
		8 $y = (x + 4)(x - 2)$

Matching graphs – cardmatch (continued)

5-8B

H 	J 	11 $y = x + 4$
U 	K 	12 $2x + y - 3 = 0$
V 	G 	13 $x + y = 3$
T 	P 	14 $x - y = 1$
Q 	S 	15 $y = x^2$
		16 $y = -x^2$
		17 $y = (x - 3)(4 - x)$
		18 $y = 4 - x^2$
		19 $y = 2^{-x}$
		20 $y = x - 2 $

Derivatives — cardmatch

5-9

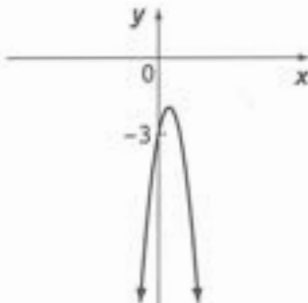
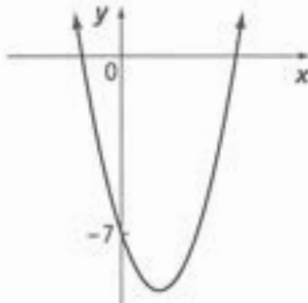
In this cardmatch activity, students are required to *match the derivative of the function* (indicated by the letters A to L) with the *original function* (indicated by the numbers 1 to 12).

$y = e^{3x}$ 1	$\frac{dy}{dx} = 3e^{3x}$ C	$y = \sin(2x + 1)$ 6
$y = xe^x$ 2	$\frac{dy}{dx} = e^x(x + 1)$ K	$\frac{dy}{dx} = 2 \cos(2x + 1)$ D
$y = \frac{e^{4x}}{3x}$ 3	$\frac{dy}{dx} = \frac{e^{4x}(4x - 1)}{3x^2}$ A	$y = \sin^2 x$ 7
$y = \log_e 3x$ 4	$\frac{dy}{dx} = \frac{1}{x}$ J	$\frac{dy}{dx} = 2 \sin x \cos x$ G
$y = x^2 \log_e x$ 5	$\frac{dy}{dx} = x(1 + 2 \log_e x)$ B	$y = \cos 3x$ 8
$y = \frac{\log_e x}{x}$ 9	$\frac{dy}{dx} = \frac{1 - \log_e x}{x^2}$ F	$\frac{dy}{dx} = -3 \sin 3x$ I
$y = \log_e (\cos x)$ 10	$\frac{dy}{dx} = -\tan x$ E	$y = e^x \tan x$ 12
$y = \frac{\tan x}{x}$ 11	$\frac{dy}{dx} = e^x (\sec^2 x + \tan x)$ L	$\frac{dy}{dx} = \frac{x \sec^2 x - \tan x}{x^2}$ H

Discriminant – cardmatch

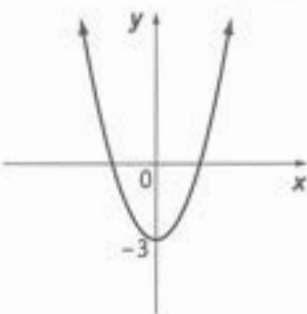
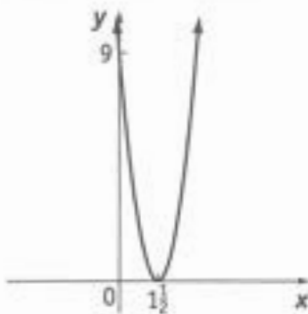
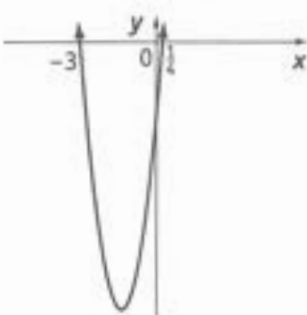
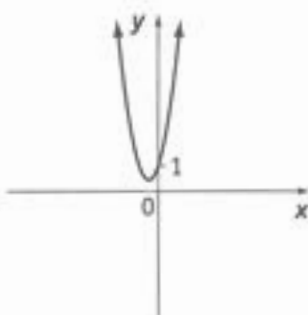
5-10A

In this cardmatch activity, students are required to match each *equation* (indicated by the letters A to F) with information about the *discriminant*, the *roots* and the *graph* of the function.

$-6x^2 + 5x - 3 = 0$	$x^2 - 3x - 7 = 0$
A	B
$\Delta = -47$	$\Delta = 37$
1	17
$\Delta < 0$	$\Delta > 0$
5	13
Has no real roots	Has two irrational roots
14	4
	Roots are $\frac{3 + \sqrt{37}}{2}$ and $\frac{3 - \sqrt{37}}{2}$
	3
	
15	20
$x^2 - 3 = 0$	$4x^2 - 12x + 9 = 0$
C	D
$\Delta = 12$	$\Delta = 0$
16	12

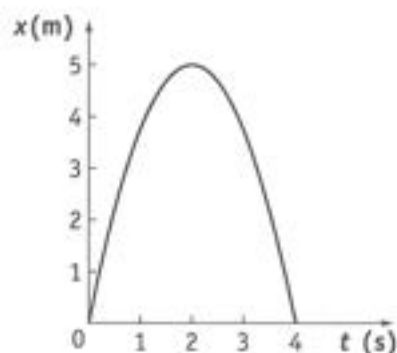
Discriminant – cardmatch (continued)

5-10B

$\Delta > 0$ 13	Has equal and rational roots 19
Has two irrational roots (which are $\sqrt{3}$ and $-\sqrt{3}$) 7	Roots are $1\frac{1}{2}$ and $1\frac{1}{2}$ 11
 2	 21
$4x^2 + 11x - 3 = 0$ E	$4x^2 + 3x + 1 = 0$ F
$\Delta = 169$ 18	$\Delta = -7$ 9
$\Delta > 0$ (and a perfect square) 6	$\Delta < 0$ 5
Has two rational roots 23	Has no real roots 14
Roots are $\frac{1}{4}$ and -3 8	
 22	 10

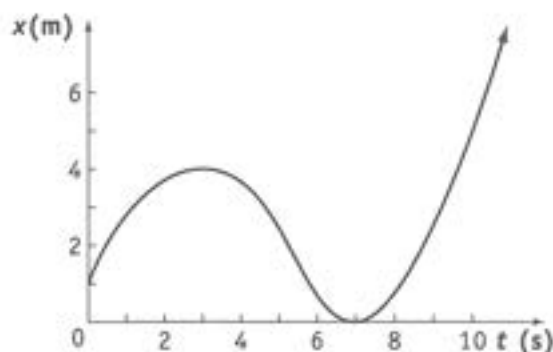
1 This diagram shows the displacement, x m, of a particle moving in a straight line at time t seconds.

- When was the particle at the origin?
- What was the maximum displacement of the particle?
- Calculate the total distance travelled in the first 4 seconds.
- Determine the average speed of the particle in the first 4 seconds.
- At what time(s) was the velocity zero?
- Sketch the graph of the particle's velocity during the first 4 seconds.
- Write a paragraph to describe this motion.



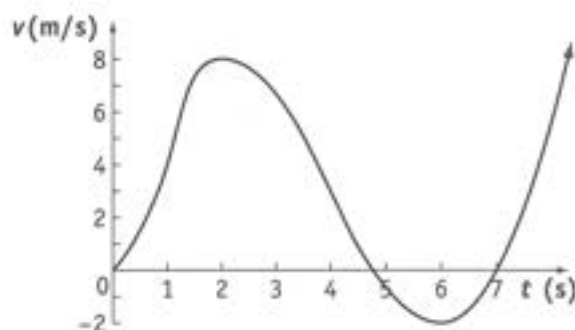
2 This diagram shows the displacement, x m, of a particle moving in a straight line at time t seconds.

- What was the initial displacement of the particle?
- When was the particle at the origin?
- During the first 10 s, what was the maximum displacement of the particle?
- Find the total distance travelled by the particle in the first 10 s.
- When was the velocity zero?
- Write a paragraph to describe this motion.



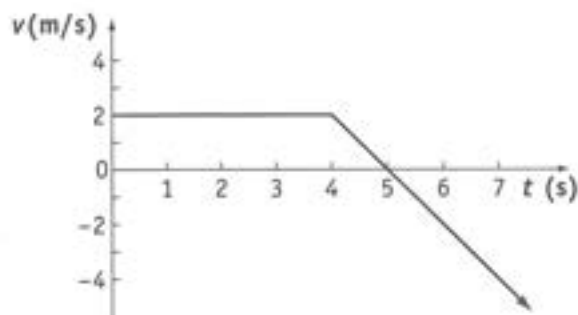
3 This diagram shows the velocity, v m/s, of a particle moving in a straight line at time t seconds.

- When was the velocity zero?
- At what times was the acceleration zero?
- What was the velocity after 6 s?
- Between what times was the acceleration negative?
- Which of these distances is the best estimate of the total distance covered by the particle in the first 4 s?
(A) 10 m (B) 16 m
(C) 20 m (D) 25 m



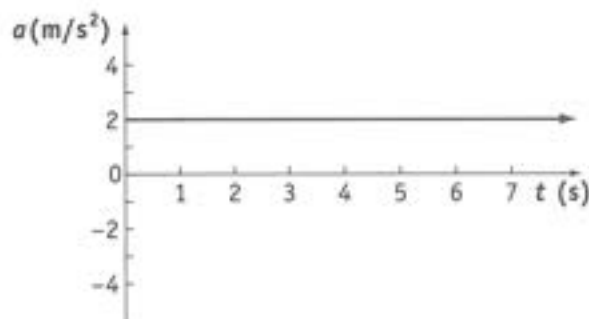
- 4** The diagram shows the velocity, v m/s, of a particle moving in a straight line at time t seconds.

- When was the particle stationary?
- Between what times was the acceleration zero?
- What was the acceleration at $t = 5$ seconds?
- How far did the particle travel during the first 7 seconds?
- How far was the particle from its starting position and in what direction was it travelling 7 seconds after the start of the motion?



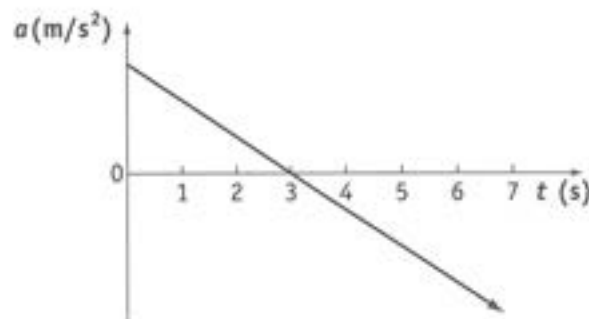
- 5** The diagram shows the acceleration, a m/s², of a particle moving in a straight line at time t seconds.

- Is it possible for the particle to change direction if its initial velocity is positive? Explain your answer.
- When the initial velocity of the particle is positive, what happens to its velocity as time increases?
- If the particle passes through the origin after 3 s, what restrictions must be placed on its initial position?



- 6** This diagram shows the acceleration, a m/s², of a particle moving in a straight line at time t seconds.

- When did the particle change direction?
 - It changed direction in the first 3 s.
 - It changed direction at 3 s.
 - It changed direction after 3 s.
 - It didn't change direction.
- At what time did the particle have the maximum positive velocity? Explain your answer.
- The particle was initially stationary 5 m to the right of the origin. In which direction, positive or negative, was its final velocity? Explain your answer.



Integration/differentiation – cardmatch

5-12

In this cardmatch activity, students are required to complete a *mixture of differentiation and integration tasks* and find the *matching answers*. Note that the constants of integration have been omitted.

1 $\frac{d}{dx} \cos \pi x$	F $\pi \sin \pi x$
2 $\int \cos \pi x \, dx$	C $\frac{1}{\pi} \sin \pi x$
3 $\frac{d}{dx} \cos \frac{x}{\pi}$	H $-\frac{1}{\pi} \sin \frac{x}{\pi}$
4 $\int \cos \frac{x}{\pi} \, dx$	A $\pi \sin \frac{x}{\pi}$
5 $\frac{d}{dx} \sin \pi x$	G $\pi \cos \pi x$
6 $\frac{d}{dx} \pi \sin x$	E $\pi \cos x$
7 $\int \pi \sin x \, dx$	K $-\pi \cos x$
8 $\int \sin \pi x \, dx$	B $-\frac{1}{\pi} \cos \pi x$
9 $\frac{d}{dx} x \sin x$	L $\sin x + x \cos x$
10 $\frac{d}{dx} x \cos x$	D $\cos x - x \sin x$
11 $\frac{d}{dx} \frac{1}{\pi} \cos x$	I $-\frac{1}{\pi} \sin x$
12 $\frac{d}{dx} \cos \frac{\pi x}{2}$	N $-\frac{\pi}{2} \sin \frac{\pi x}{2}$
13 $\frac{d}{dx} \sin \frac{\pi x}{2}$	J $\frac{\pi}{2} \cos \frac{\pi x}{2}$
14 $\int \sin \frac{\pi x}{2} \, dx$	O $-\frac{2}{\pi} \cos \frac{\pi x}{2}$
15 $\int \cos \frac{\pi x}{2} \, dx$	M $\frac{2}{\pi} \sin \frac{\pi x}{2}$

These four questions are based on geometric series.

1 Win Lotto

Janina borrowed \$195 000 to buy a house. She is repaying the loan, plus interest, in monthly instalments of \$1800 at 9% pa, monthly reducible interest.

On the same day that Janina made her 21st monthly repayment she won Lotto and decided to repay the balance of her loan. How much will she owe on her loan immediately after she makes her 21st repayment?

2 Annuities

55-year-old Fred is about to retire. He is trying to decide what to do with \$200 000 of his superannuation. He has 2 options:

Option 1: Buy a whole of life pension (called an allocated pension), which will pay him \$2200 each month for the rest of his life

or

Option 2: Invest the money himself. If he invests the money himself he can earn an interest rate of 6% pa compounding monthly (after tax). From this investment, Fred will withdraw \$2200 per month.

How long will he need to live so that he is better off taking the pension option rather than investing the money himself? Answer correct to the nearest year.

3 Hospital maths

Wilma is a terminally ill patient in hospital. Every 4 hours she is given a 5 mg injection of pain-killer. During the following 4 hours, 90% of the total amount of pain-killer in her body is eliminated.

Over an extended period, what is the maximum amount of pain-killer that could be in Wilma's blood at any one time? Express your answer correct to the nearest thousandth of a milligram.

4 Loan repayments

Richard borrowed \$100 000 to buy an apartment, which he is going to rent. He plans to repay \$500 every fortnight from his salary. On every 4th week he will repay an extra \$800 from the rent he receives. Interest on the loan is 6.5% pa reducing fortnightly.

Let A_n = the amount owing on the loan after n fortnights.

(a) Show that the interest rate of 6.5% pa is equivalent to 0.25% per fortnight.

(b) Show that $A_2 = 100\,000R^2 - 500(R + 1) - 800$ where $R = 1.0025$.

(c) Show that $A_4 = 100\,000R^4 - 500(R^3 + R^2 + R + 1) - 800(R^2 + 1)$.

(d) Hence show that A_{100} can be calculated by evaluating:

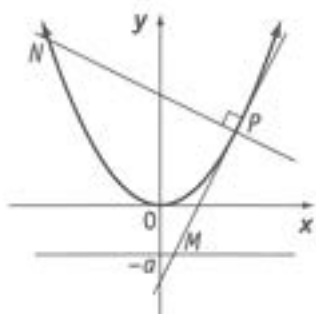
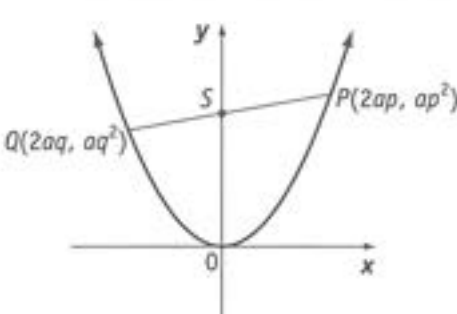
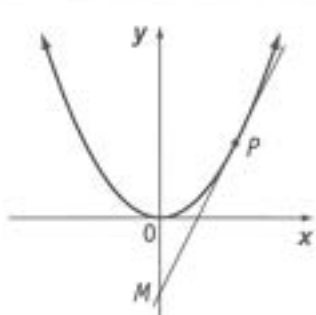
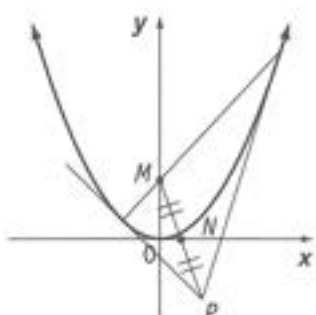
$$A_{100} = 100\,000R^{100} - 500 \times \frac{R^{100} - 1}{R - 1} - 800 \times \frac{R^{100} - 1}{R^2 - 1}$$

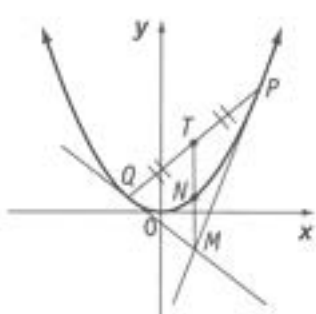
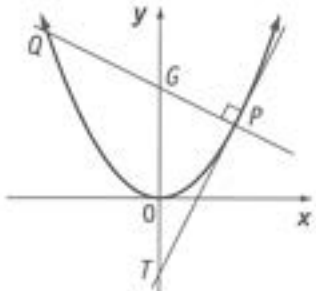
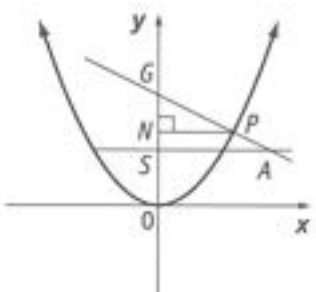
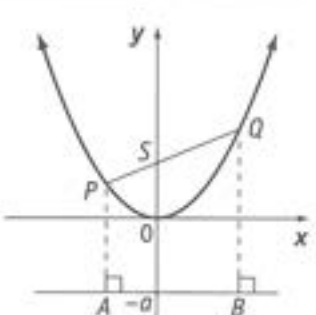
(e) What percentage of the money repaid in the first 100 repayments will be interest?

Parabola language – cardmatch

5-14A

In this cardmatch activity, students are required to match a *word description* with a *diagram* showing the same information.

Language	Diagram
The tangent at $P(2ap, ap^2)$ on the parabola $x^2 = 4ay$ cuts the directrix at M. The normal at P meets the parabola again at N. A	 2
The chord joining the points $t = p$ and $t = q$ on the parabola $x = 2at, y = at^2$ passes through the focus. B	 4
The tangent at $P(2ap, ap^2)$ on the parabola $x^2 = 4ay$ cuts the axis of symmetry at M. C	 1
P is an external point to the parabola $x^2 = 4ay$. The chord of contact of the tangents drawn from P cuts the axis of symmetry at M. The midpoint of PM is shown as point N. D	 6

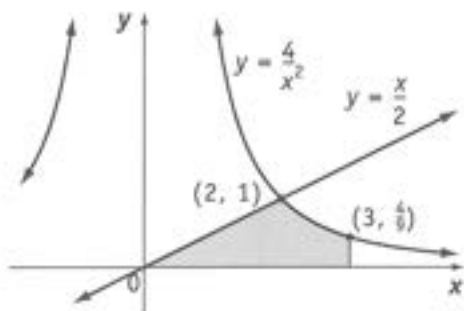
Language	Diagram
P and Q are points on the parabola $x^2 = 4ay$. The tangents at P and Q meet at M, and T is the midpoint of PQ. MT cuts the parabola at N. E	 7
At the point $P(2ap, ap^2)$ on the parabola $x^2 = 4ay$ a tangent is drawn to cut the axis of the parabola at T. The normal from P meets the axis of symmetry at G. F	 3
The normal at $P(2ap, ap^2)$ on $x^2 = 4ay$ meets the axis of symmetry at G and the latus rectum produced at A. PN is perpendicular to the axis of symmetry. G	 5
PQ is a focal chord to $x^2 = 4ay$. Perpendiculars from P and Q meet the directrix at A and B respectively. H	 8

1 Find the equation of the parabola with:

- vertex (1, 4) and focus (1, 7)
- focus (3, 0) and directrix $x = -1$

2 Find the size of the area enclosed between the curve $y = x^2 - 4x$ and the line $y = 2 - 3x$.

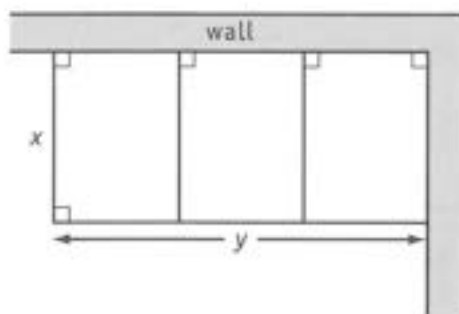
3 Calculate the size of the shaded area in this diagram.



4 Find the equation of the tangent to $y = x\sqrt{x-1}$ at the point (5, 10).

5 Hassan is having trouble with his horses. He has decided that he has to build separate feeding yards for them.

He has 72 m of fencing to use and he is going to build 3 feeding pens, using the walls in the corner of the barn yard as 2 sides of the pens, as shown in the diagram.



(a) Show that the area, $A \text{ m}^2$, he can contain inside the 3 pens is given by $A = 72x - 3x^2$.

(b) Hence show that the area will be the maximum when he places half of the fencing parallel to the longer section of barn wall.

6 Prove that $(1 - \tan x)^2 + (1 + \tan x)^2 = 2 \sec^2 x$.

7 Solve :

(a) $a^2 - 4 > 3a$

(b) $|2b - 4| < 6$

(c) $-2 \leq \frac{3x-1}{4} < 5$

(d) $3 - \frac{2x-1}{4} = 5$

1 Evaluate, correct to 3 decimal places:

(a) $\sin 1$

(b) $\cos 0.5$

2 Solve the following equations for $0 \leq \theta \leq 2\pi$.

(a) $2 \cos \theta - 1 = 0$

(b) $\tan^2 \theta = 3$

(c) $\sqrt{2} \sin 2\theta = 1$

3 On the same axes sketch the following trigonometric functions for $0 \leq x \leq 2\pi$.

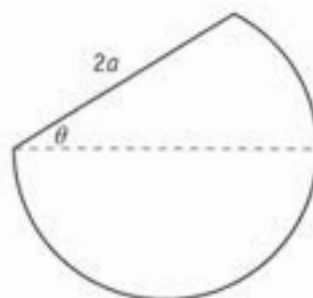
(a) $y = \sin x$

(b) $y = \cos x$

(c) $y = 3 \sin x$

4 What is the equation of the parabola with vertex $(2, 3)$ and focus $(0, 3)$?

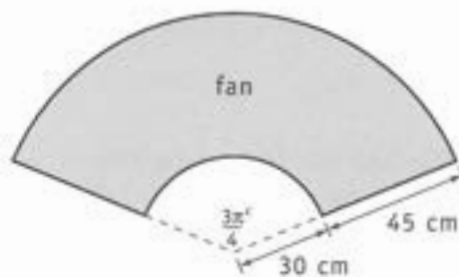
5 This diagram contains a semicircle surmounted by a sector. Derive expressions for the perimeter and the area of the diagram.



6 The diagram shows a traditional Asian ceremonial fan. Calculate the exact:

(a) shaded area of the fan

(b) perimeter of the shaded fan section



7 The cost of running a ship at a constant speed of V km/h is given by the expression $160 + \frac{1}{100}V^3$ dollars per hour.

(a) How long will a journey of 1000 km take at a constant speed of 10 km/h?

(b) Find the cost of a 1000 km journey at a speed of 10 km/h.

(c) Find the cost, \$C, of a 1000 km trip at a constant speed of V km/h.

(d) Find the most economical speed for the journey and hence the minimum cost for the 1000 km trip.

(e) If the ship's maximum speed is 16 km/h, what is the minimum cost of a 1000 km trip?

1 Differentiate:

(a) $y = \cos 2x$

(b) $y = \sin^3 x$

(c) $y = \frac{\sin 4x}{x}$

2 Find the locus of:

(a) the points that are equidistant from $A(8, 2)$ and $B(6, 4)$

(b) the points that are equidistant from the line $y = 6$ and the point $(0, 0)$

(c) the points that are twice as far from $(0, 2)$ as they are from $(0, 5)$

3 (a) Show that when x is restricted to $-\pi \leq x \leq \pi$, there is only one real solution to the equation $\sin x = -2x$.

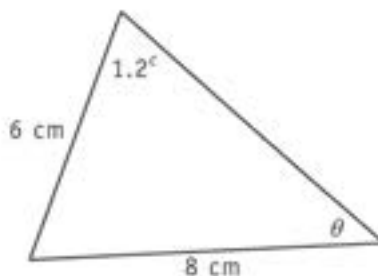
(b) What is the maximum number of real solutions that the equation $\sin x = -2x$ can have when the restriction on x is $-2\pi \leq x \leq 2\pi$?

4 Solve $\sec^2 x = 2$ for $0 \leq x \leq 2\pi$.

5 Use the trapezoidal rule with 4 function values to find the approximate value of $\int_0^1 \tan x \, dx$.
(Note: Use radians!)

6 Use Simpson's rule with 3 function values to find the approximate value of $\int_0^{\frac{\pi}{3}} \sin x \, dx$.
Answer correct to 4 decimal places.

7 Use the sine rule to find the size of θ .
Answer in radians correct to 1 decimal place.



8 Prove that:

(a) $\sin^2 \theta \operatorname{cosec} \theta + \cos^2 \theta \sec \theta = \sin \theta + \cos \theta$

(b) $\frac{\sin^3 \theta}{\cos \theta} + \sin \theta \cos \theta = \tan \theta$

(c) $\sqrt{1 - \sin^2 \theta} \times \tan \theta = \sin \theta$

9 (a) Solve the inequality $x^2 + 2x - 3 \geq 0$.

(b) Find the values of m for which the quadratic equation $x^2 - (m - 3)x - (2m - 3) = 0$ has real roots.

- 1 Find the expression for $\frac{dy}{dx}$ for:
- (a) $y = x \sin 2x$ (b) $y = \frac{3x}{\cos x}$ (c) $y = \tan^2 x$

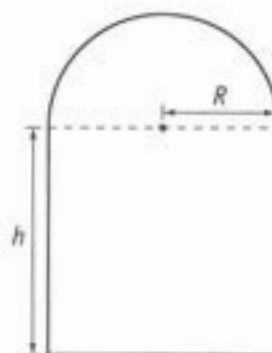
- 2 Find the equation of the tangent to $y = 4 \sin x$ at the point $(\frac{\pi}{6}, 2)$.

- 3 At what point(s) on the curve $y = \cos x$ is the tangent parallel to the line $2y = x$?
Answer for $0 \leq x \leq 2\pi$.

- 4 Use Simpson's rule with 5 function values to approximate $\int_0^1 \sec x \, dx$.

- 5 For the curve $y = 2x^3 - 3x^2 - 12x$:
- (a) Find the coordinates of the turning points.
(b) Sketch the curve in the domain $-2 \leq x \leq 3$.
(c) State the range of the curve in $-2 \leq x \leq 3$.

- 6 A window frame is in the shape of a rectangle surmounted by a semicircle, as shown in the diagram. The perimeter of the window is 240 cm. Let the radius of the semicircle be R cm.



- (a) Show that $h = \frac{240 - \pi R - 2R}{2}$.
- (b) Hence show that the area, $A \text{ cm}^2$, of the window is given by $A = 240R - \frac{1}{2}R^2(\pi + 4)$.
- (c) Hence show that for a maximum area, $h = R$ (i.e. the height is the same as the radius of the semicircle).
- 7 (a) Differentiate $\frac{1}{x^2}$.
- (b) Show that the derivative of $\frac{1}{\sin x}$ can be written as $-\cot x \operatorname{cosec} x$.

1 Find $\frac{dy}{dx}$ for:

(a) $y = \sqrt{4x + 2}$

(b) $y = \sin x \cos x$

(c) $y = x \cos 4x$

2 Find:

(a) $\int (3x - 1)^2 dx$

(b) $\int (x^2 - 1)^2 dx$

(c) $\int \frac{10}{x^2} dx$

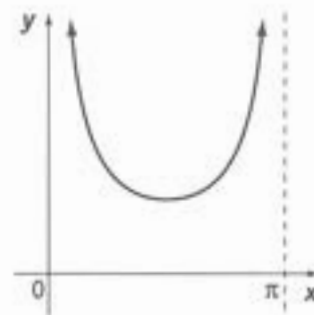
3 The diagram shows the graph of the section of $y = \operatorname{cosec} x$ between $x = 0$ and $x = \pi$.

(a) Explain why it is not possible to

evaluate $\int_0^\pi \operatorname{cosec} x dx$.

(b) Use the trapezoidal rule with 6 function

values to approximate $\int_{0.1}^{0.6} \operatorname{cosec} x dx$.



4 Find the volume generated when the section of $y = x^2 - 4$ between $(2, 0)$ and $(3, 5)$ is rotated about:

(a) the x -axis

(b) the y -axis

5 (a) Sketch the graph of $y = 4 \sin 2x$ for $-\pi \leq x \leq \pi$.

(b) What are the period and amplitude of $y = 4 \sin 2x$?

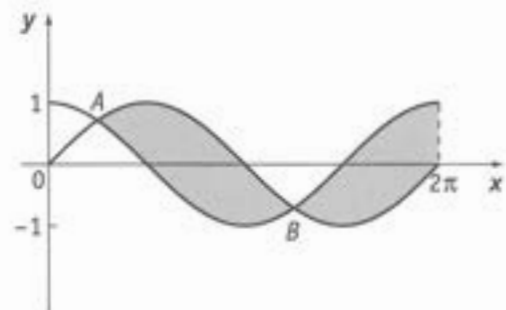
(c) On your graph show the approximate positions of the points on the curve where the gradient of the curve is 1. How many places are there?

(d) Find the x -coordinates of the points on $y = 4 \sin 2x$ where the gradient is 1. Answer in radians correct to 2 decimal places.

6 This diagram shows the graphs of $y = \sin x$ and $y = \cos x$ for $0 \leq x \leq 2\pi$.

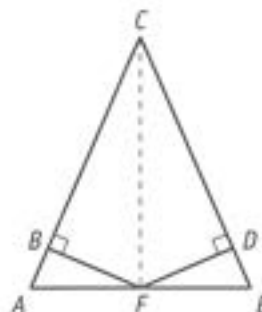
(a) Find the x -coordinates of the points labelled A and B .

(b) Find the total area of the 2 shaded sections in the diagram.

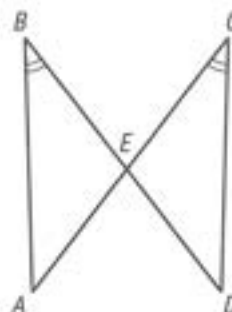


- 1 $\triangle ACE$ is isosceles with $AC = EC$.
 F is the midpoint of AE , $\angle ABF = 90^\circ$
 and $\angle FDE = 90^\circ$.

- (a) Prove that $\triangle ABF \cong \triangle EDF$.
 (b) Hence show that $BF = DF$.



- 2 In the diagram, $\angle ABE = \angle DCE$.
 (a) Prove that $\triangle ABE \cong \triangle DCE$.
 (b) Hence find the length of BE , given that
 $CE = 15$ cm, $AE = 18$ cm and $ED = 12$ cm.



- 3 Simplify the following expressions. Answer in simplest index form.

(a) $3^n \times 9^n \times 27^{2n}$ (b) $\frac{12^{2n} \times 6^2}{18^n}$

- 4 Find the vertex, focus and directrix of the parabola $12y = x^2 - 4x$.

- 5 Prove that the quadratic equation $x^2 - (3b + 2c)x + 3bc = 0$ always has real roots, for all real values of b and c .

- 6 One root of the quadratic equation $x^2 + (1 - 2k)x + 6k = 0$ is 1 more than the other root. Find the values of k and the 2 roots.

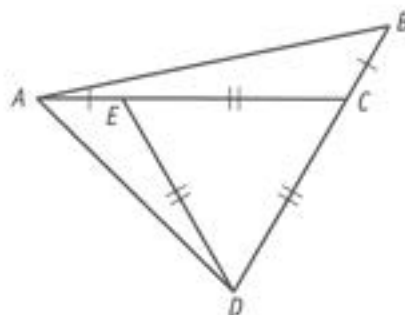
- 7 Use Simpson's rule with 5 function values to approximate $\int_0^1 x \sin x \, dx$.

- 8 The roots of the quadratic equation $2x^2 - 6x + 1 = 0$ are α and β . Find the value of:

(a) $\alpha + \beta$ (b) $\alpha\beta$ (c) $\frac{1}{\alpha} + \frac{1}{\beta}$
 (d) $\alpha^2\beta + \alpha\beta^2$ (e) $\alpha^2 + \beta^2$

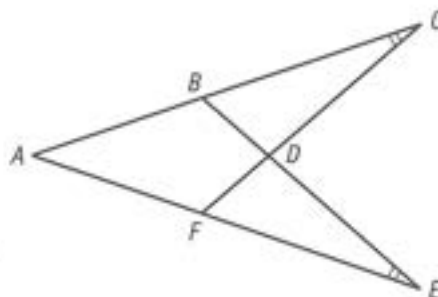
1 Given that $AE = BC$ and $ED = EC = CD$:

- Prove that $AD = BE$.
- Prove that $\angle ADC = \angle BED$.



2 Given that $\angle ACF = \angle AEB$:

- Prove that $\triangle CAF \parallel \triangle EAB$.
- Given that $AB = 9$ cm, $AF = 12$ cm and $FE = 8$ cm, determine the length of AC .



3 The quadratic equation $x^2 + (m - 3)x + (m - 1) = 0$ has one root that is half the other. Find the value of m and the values of the two roots.

4 Find the relationship between the values of a and c in the quadratic equation $ax^2 + 5x + c = 0$, given that the roots are reciprocals of each other.

5 $P(-\frac{4}{3}, \frac{2}{9})$ and $Q(12, 18)$ are points on the parabola $x^2 = 8y$.

- Show that the line through P and Q also passes through the focus of the parabola.
- Find the equations of the tangents to the parabola at P and Q .
- Show that the tangents at P and Q intersect:
 - at right angles
 - on the directrix of the parabola

6 (a) Show that $\frac{1}{\alpha - 1} + \frac{1}{\beta - 1} = \frac{(\alpha + \beta) - 2}{\alpha\beta - (\alpha + \beta) + 1}$.

(b) α and β are the roots of $x^2 + 10x - 7 = 0$. Find the value of $\frac{1}{\alpha - 1} + \frac{1}{\beta - 1}$.

1 Simplify $\frac{3^{n+1} \times 9^{n-1}}{27^{1-n}}$.

2 For what value of x is:

(a) $3 \log_2 x = 12$?

(b) $2 \log_a x = \log_a 9$?

(c) $\log_3 (x+1) - \log_3 (x-1) = 2$?

3 What is the value of $\log_3 24$ correct to 2 decimal places?

4 (a) Show that the graph of $y = 3x \log_e x$ has only one stationary point and determine its coordinates and nature.

(b) For what values of x is the expression $x \log_e x$ defined?

(c) Hence sketch the function $y = 3x \log_e x$.

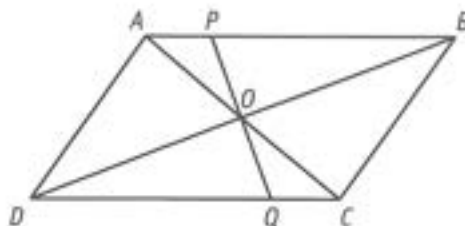
(d) What is the range of the expression $3x \log_e x$?

5 Simplify $\frac{5}{x+2} - \frac{1}{x^2-4}$.

6 $ABCD$ is a parallelogram with diagonals intersecting at point O . The interval PQ also passes through point O .

(a) Explain how you know that $AO = OC$.

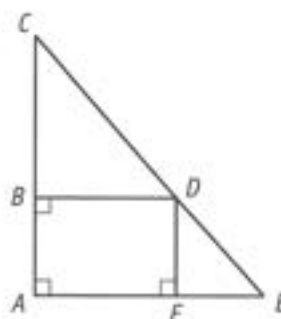
(b) Prove that a suitable pair of triangles are congruent and hence prove that $PO = OQ$.



7 $\triangle ACE$ is a right-angled triangle and $ABDF$ is a rectangle. Point D lies on CE . $CD = x$ and $DE = y$.

(a) Prove that $AE = \left(1 + \frac{y}{x}\right) BD$.

(b) Hence prove that $EF = \frac{y}{x} \times BD$.



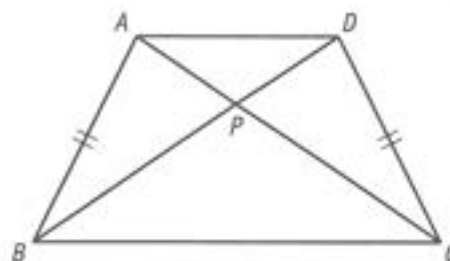
8 Solve the equation $25e^{2t-1} = 7500$. Answer correct to 3 significant figures.

9 Solve the inequality $4x^2 > 20x$.

10 For what values of θ does $y = 2 \cos \theta + 1$ cross the x -axis in the domain $0 \leq \theta \leq 2\pi$?

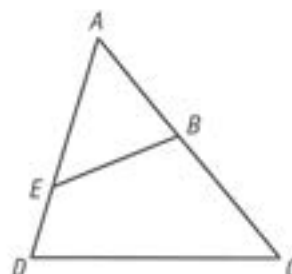
1 $ABCD$ is an isosceles trapezium.

- Explain how you know that $\angle DCB = \angle ABC$.
- Prove that $\triangle DAC = \triangle ADB$.
- Hence, or otherwise, prove that $\triangle CPB$ is isosceles.



2 Given that $\angle AEB = \angle ACD$:

- Prove that $\triangle AEB$ is similar to $\triangle ACD$.
- Given that $AE = 24$ cm, $ED = 8$ cm and $AB = 20$ cm, find the lengths of AC and BC .



- Find the equation of the tangent to $y = x \log_e x$ at the point $(1, 0)$.
- Find the area enclosed by the tangent and the coordinate axes.

4 Solve the equation $2 \log_a x = \log_a (x + 6)$.

5 Simplify $\log_e e^2 - \log_e \sqrt{e} + \log_e e$.

- On one set of axes show the graphs of $y = \sin 2x$ and $y = 2 \cos x$ for $0 \leq x \leq 2\pi$.
- Write down the period and amplitude of both graphs.
- Determine the size of the area enclosed between the two curves between the x -values $\frac{\pi}{2}$ and $\frac{3\pi}{2}$.

7 Find all the solutions in $0 \leq \theta \leq 2\pi$ to these equations:

- $2 \cos 2\theta - \sqrt{3} = 0$
- $\sin \theta = \sqrt{3} \cos \theta$

8 Use the trapezoidal rule with 4 function values to find the approximate value of $\int_{0.6}^{1.2} \tan x \, dx$. Answer correct to 2 significant figures.

9 Simplify these expressions, leaving your answers in index form:

- $3^x \times 3^x$
- $3^x + 3^x + 3^x$
- $(3^x)^2$
- $\sqrt{9^{16x}}$

1 Sketch the following functions and state the domain and range of each function:

(a) $y = 10 - e^{-x}$

(b) $y = \log_e 2x$

(c) $y = 1 + 2e^x$

2 At what point on the curve $y = \log_e 2x$ is the gradient equal to 3?

3 (a) Show that $\frac{5x-1}{(3x+1)(x-1)} = \frac{2}{3x+1} + \frac{1}{x-1}$.

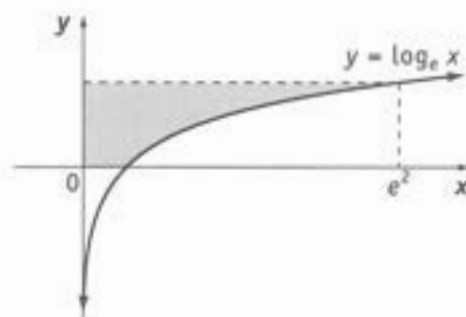
(b) Hence show that $\int \frac{5x-1}{(3x+1)(x-1)} dx = \log_e [(x-1)(3x+1)^{\frac{2}{3}}] + C$.

4 The diagram shows the graph of $y = \log_e x$.

(a) Show that the shaded area can be

found by evaluating $\int_0^2 e^y dy$.

(b) Hence find the shaded area.



5 (a) Show that the curve $y = xe^{2x}$ only has one stationary point.

(b) Determine the coordinates and nature of this stationary point.

(c) Hence sketch the curve $y = xe^{2x}$.

(d) What is the range of the function $y = xe^{2x}$?

6 $\int_0^K 6e^{2x} dx = 3(e^7 - 1)$. What is the value of K ?

7 Find the volume generated when the section of $y = e^x + 1$ between $x = 1$ and $x = 3$ is rotated about the x -axis?

8 Solve the equation $\log_2 (3x + 1) - 4 = \log_2 (x - 1)$.

9 Differentiate $y = \sin^3 x$ and hence find the primitive of $\sin^2 x \cos x$.
[Remember! $\sin^3 x = (\sin x)^3$]

10 (a) Show that $\frac{d}{dx} x \log_e x = 1 + \log_e x$

(b) Hence find $\int_1^e (\log_e x) dx$.

Parabola revision worksheet

1 Write down the vertex, focus and directrix of each parabola:

(a) $(x - 1)^2 = 12(y + 2)$

(b) $(x + 3)^2 = -18(y - 4)$

(c) $y^2 = 8(x - 1)$

(d) $(y - 2)^2 = -2(x - 1)$

(e) $x^2 = 8(y + 5)$

(f) $y = x^2 - 12$

2 Find the coordinates of the focus and the equation of the directrix of each parabola:

(a) $x^2 - 10x - 24y = 47$

(b) $y^2 + 4y + 4x = 0$

(c) $x^2 + 6x - 6y = 0$

3 Write down the equation of each parabola, given the following features:

(a) vertex (2, 5) and focus (2, 9)

(b) vertex (3, 6) and focus (3, 0)

(c) vertex (2, 5) and focus (3, 5)

(d) focus (0, 5) and directrix $x = 6$

(e) focus (0, 4) and directrix $y = 8$

4 Find the equation of the tangent to the parabola $x^2 = 8y$ at the point $(2, \frac{1}{2})$.

5 Find the equation of the normal to $x^2 = -4y$ at the point (4, -16).

6 Determine the equation of the locus of all the points that are equidistant from (0, 10) and the x -axis.

7 (a) Show that $y = 2x - 3$ is a tangent to the parabola $x^2 = 4(y - 1)$.

(b) At what point does the tangent touch the parabola?

8 (a) For the parabola $x^2 = 8y - 24$, find the vertex and show that the focus is (0, 5).

(b) Sketch the parabola and find the equation of the line (L) through the focus and parallel to the directrix.

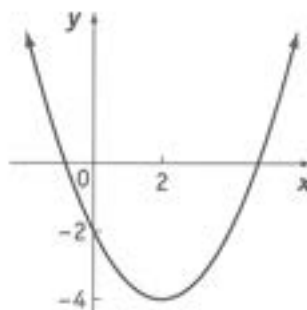
(c) Find the points of intersection of line L and the parabola.

(d) Calculate the area enclosed by the parabola and line L .

9 The diagram shows a parabola with vertex (2, -4) and y -intercept -2.

(a) Find the equation of the parabola.

(b) What are the coordinates of the focus?



Holiday assignment

- 1** Calculate the following indefinite integrals:

(a) $\int (3x^2 - 5) dx$

(b) $\int \frac{4x^4 + 6x^2 + 5x}{x} dx$

(c) $\int (3x + 4)^5 dx$

(d) $\int \sqrt{6x + 7} dx$

- 2** Evaluate the definite integral $\int_2^5 (3x^2 - 6) dx$.

- 3** Find the area between the curve $y = x^2 + 2$, the x -axis and the lines $x = 1$ and $x = 5$.

- 4** Calculate the volume of the solid formed by rotating the section of the line $y = 2x + 1$ between $x = 0$ and $x = 3$ about the x -axis.

- 5** (a) Copy and complete this table of values for $y = \sqrt{4 - x^2}$.
Express each value correct to 3 decimal places.

x	0	0.5	1	1.5	2
y					

- (b) By using Simpson's rule with 5 function values, find the approximate value of $\int_0^2 \sqrt{4 - x^2} dx$.

- (c) Write a sentence to describe how you could check the accuracy of this approximation.

- 6** (a) Express A in terms of B , given that $\frac{dA}{dB} = 6B^{-3}$ and $A = 10$ when $B = 1$.

- (b) Given $f'(x) = 4x - 1$ and $f(2) = 3$, find $f(x)$.

- 7** (a) Find the area enclosed by the curve $y = x^2 + 1$ and the line $y = 2x + 1$.

- (b) Calculate the area between the curve $y = 4 - x^2$ and the x -axis between $x = 1$ and $x = 3$.

- 8** Consider the curve given by $y = 3x^4 + 8x^3$.

- (a) Find the coordinates of any stationary points.

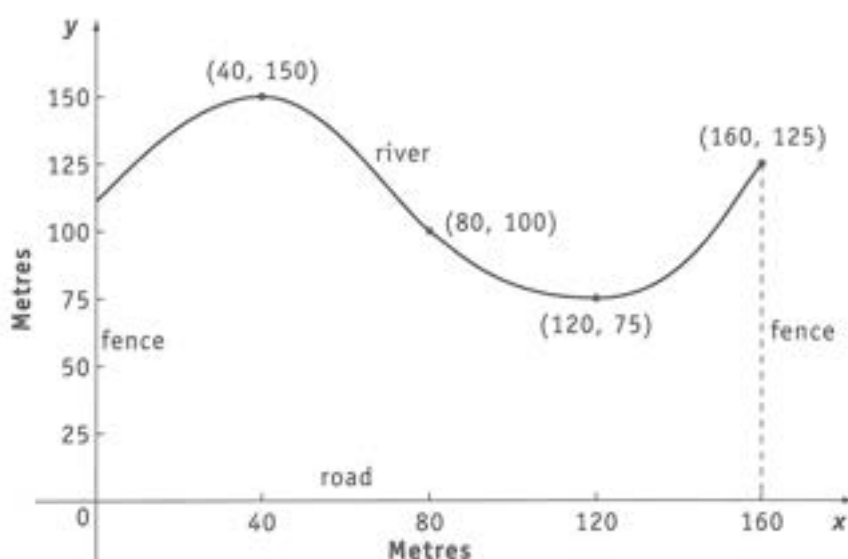
- (b) Determine the nature of the stationary points.

- (c) Find the coordinates of any points of inflection.

- (d) Sketch the curve.

Holiday assignment

9



This diagram shows a paddock bounded by a river, a road and 2 fences perpendicular to the road. Farmer Ben wishes to buy the paddock. However, before he buys it he wants to check the area of this paddock. Every 40 m Ben has measured the perpendicular distance from the road to the river. These distances are shown on the diagram.

- (a) Using the road as the x -axis and the fences as the y -axis and the line $x = 160$, copy and complete this table of values:

x	0	40	80	120	160
y	110				

- (b) Estimate the area of the paddock using the trapezoidal rule with 5 function values.

- 10** (a) In the equation $2x^2 + (k + 1)x + 4k = 0$, the sum of the roots equals the product of the roots. Find the value of k .

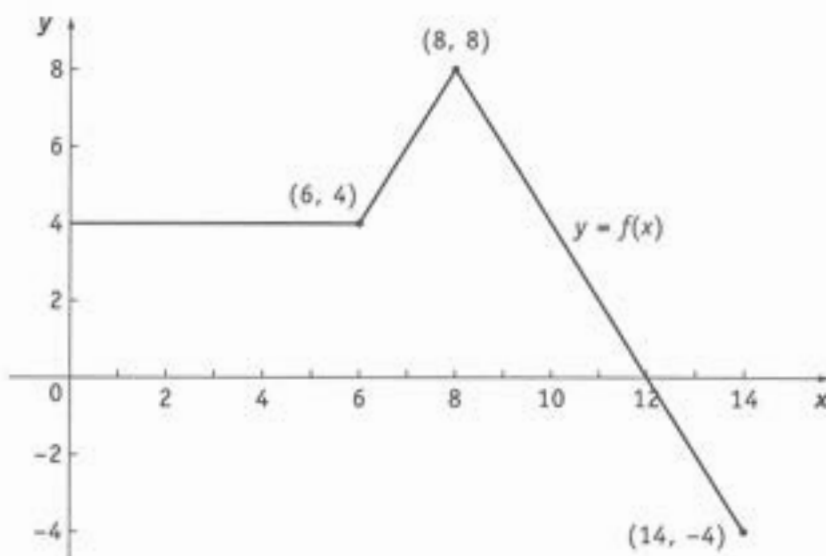
- (b) For what values of m does the equation $x^2 + mx + 9 = 0$ have no roots?

- 11** In the equation $kx^2 - (k + 2)x + 2k - 1 = 0$, what is the value of k when:

- (a) the roots are reciprocals of each other?
 (b) the sum of the roots equals twice the product of the roots?

Holiday assignment

12



This diagram shows the graph of $y = f(x)$.

Use any suitable method to find the *exact* value of $\int_0^{14} f(x) dx$.

13 For the parabola $(x - 4)^2 = -20(y + 1)$, find the:

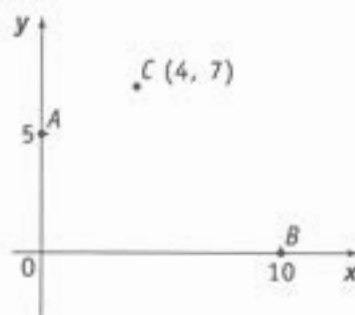
- coordinates of the vertex and the focus
- equation of the directrix
- equation of the axis of symmetry

14 The equation of a parabola is $y^2 - 8y + 10x + 1 = 0$. Find the:

- coordinates of the vertex and the focus
- equation of the directrix

15 The points $A(0, 5)$, $B(10, 0)$ and $C(4, 7)$ are shown on this number plane.

- Calculate the distance AC .
- Find the gradient of the line AB .
- Find the gradient of the line BD where the line BD is perpendicular to the line AB .
- Show that the equation of the line AC is $2y = x + 10$.
- The line AC cuts the x -axis at F . Find the coordinates of F .
- Prove that $\triangle AOF = \triangle AOB$.



Mixed revision

1 Differentiate:

(a) $y = \sin 2x$

(b) $y = \tan \frac{x}{2}$

(c) $y = x^2 - \cos \pi x$

(d) $y = 3x + e^{-2x}$

(e) $y = 3x \log_e (x^2 - 7)$

(f) $y = \frac{4}{e^x}$

(g) $y = (2 - 3x)^{10}$

(h) $y = \frac{x^2}{\sin x}$

(i) $y = \pi x - e$

2 Find an expression for each of the following indefinite integrals:

(a) $\int (2x + e^{3x}) dx$

(b) $\int 2\pi \sin 3x dx$

(c) $\int \sec^2 \frac{x}{3} dx$

(d) $\int \frac{3x}{x^2 + 5} dx$

(e) $\int x \sqrt{x} dx$

(f) $\int \left(\frac{1}{x} + e^{-x} \right) dx$

3 Solve the equation $12e^{2x} = 750$. Express your answer correct to 2 decimal places.

4 (a) Show that $\frac{2x+1}{x-1} = 2 + \frac{3}{x-1}$.

(b) Hence evaluate $\int_2^5 \frac{2x+1}{x-1} dx$.

5 Show that $e + 1 + e^{-1} + e^{-2} + e^{-3} + e^{-4} + \dots = \frac{e^2}{e-1}$.

6 (a) Show that the equation of the tangent to the curve $y = x^2 - 4x$ at the point where $x = 1$ is $2x + y + 1 = 0$.

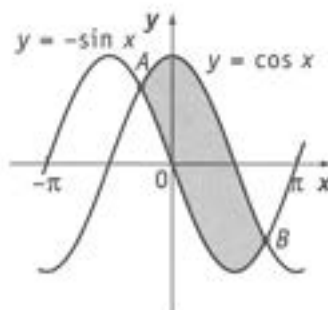
(b) Determine the coordinates of the points where the tangent $2x + y + 1 = 0$ cuts the coordinate axes.

(c) Hence find the area enclosed between the coordinate axes and the tangent.

7 The diagram shows the graphs of $y = \cos x$ and $y = -\sin x$ for $-\pi < x < \pi$.

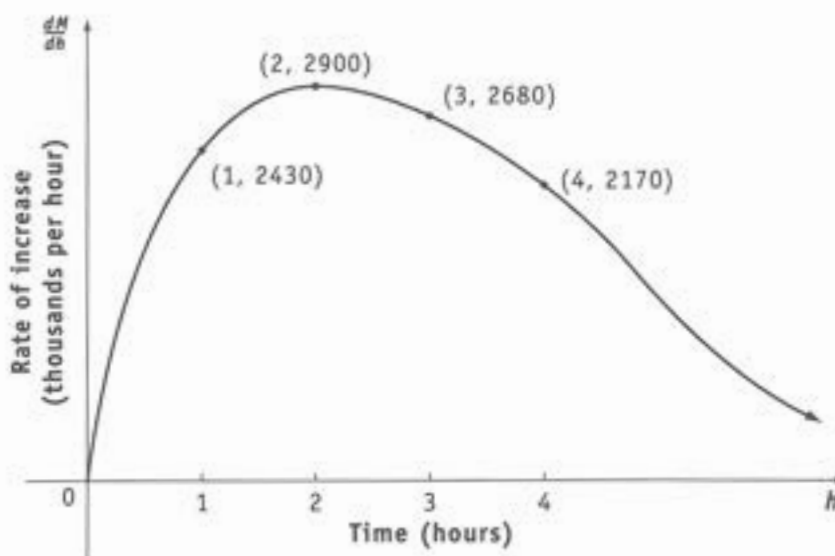
(a) Determine the coordinates of points A and B.

(b) Find the exact value of the size of the shaded area.



Mixed revision

- 8 (a) Determine the coordinates of the point(s) where the curve $y = 2x^2e^{2x}$ crosses the coordinate axes.
 (b) Differentiate the equation $y = 2x^2e^{2x}$.
 (c) Hence find the coordinates of the stationary points on the curve $y = 2x^2e^{2x}$, and determine their nature.
 (d) Describe what happens to the curve $y = 2x^2e^{2x}$ for very large positive and negative values of x .
 (e) Hence sketch the graph of $y = 2x^2e^{2x}$. Label the stationary points.
- 9 (a) Show that the equation $y = \log_e(x - 1)$ can be rewritten as $x = 1 + e^y$.
 (b) Hence find the size of the area enclosed by the curve $y = \log_e(x - 1)$, the x -axis and the line $x = 5$.
- 10 (a) Show that $M = M_0e^{kt}$ is a solution to the differential equation $\frac{dM}{dt} = kM$.
 (b) Hence or otherwise, find the equation of the curve that passes through the point $(0, 4)$ and whose gradient function is given by $\frac{dy}{dx} = 2y$.
- 11 The number of bacteria (B) in an untreated infection is increasing. The diagram shows the birth rate of bacteria (h , in thousands per hour). The equation of the rate of increase is $\frac{dB}{dh} = 4he^{-0.5h}$.



- (a) Use the trapezoidal rule with 4 function values to estimate the total number of bacteria born in the first 3 hours.
 (b) Sketch a curve to represent the number of bacteria present in the first h hours of the infection.

More mixed revision

- 1 Solve $5 \times 2^{2002} + 2^{2001}x - 2^{2000} = 2^{1999}$.
- 2 Solve:
 - (a) $x^2 - x \geq 20$
 - (b) $\frac{x+2}{x} > 3$ (where x is a positive number)
 - (c) $|2x+1| < 9$
- 3 α and β are the roots of the quadratic equation $4x^2 - 3x - 2 = 0$. Without first finding the values of α and β , determine the value of:
 - (a) $\alpha^3 + \beta^3$
 - (b) $\frac{5}{\alpha} + \frac{5}{\beta}$
 - (c) $\alpha^2\beta + \alpha\beta^2$
- 4 For what value of K will the equation $2x^2 + 5x + K = 0$ have equal roots?
- 5 Gareth currently has \$100 000 in his superannuation fund, which pays 0.7% interest per month. When he retires in 8 years time, he will need \$350 000. How much will he need to invest at the end of each month for the next 8 years so that he will have \$350 000 when he retires?
- 6 (a) Show that $\frac{d}{dx} \frac{1}{4}e^{2x} (2x^2 - 2x + 1) = x^2e^{2x}$.
 (b) Hence, or otherwise, find the volume generated when the section of the curve $y = xe^x$ that is between $x = 0$ and $x = 1$ is rotated about the x -axis.
- 7 The volume, $V \text{ m}^3$, of water in a dam t days after the start of a drought is given by $V = 450 \times e^{-\frac{t}{20}}$.
 - (a) How many cubic metres of water were in the dam at the beginning of the drought?
 - (b) How long did it take for half of the water to be used?
 - (c) At what rate was the water being used 10 days after the start of the drought?
- 8 (a) Solve $2y^2 + 5y - 3 = 0$.
 (b) Hence, or otherwise, find all the solutions to $2 \sin^2 x + 5 \sin x - 3 = 0$ in $0 \leq x \leq 4\pi$.
 (c) The velocity of a particle at t seconds is given by $V = 2 \sin^2 t + 5 \sin t - 3$ m/s.
 For what values of t is the particle stationary in $0 \leq t \leq 12.6$?

5-1 Factorisation classification — cardmatch

- 1 D, F, H, I, P, R, U, V
 2 B, E, M, T, Y
 3 J, N, X, CC, GG, II
 4 C, G, Z, AA, BB, DD, FF, JJ
 5 K, Q, W, EE, HH
 6 A, L, O, S
 A $(2x - 1)(4x^2 + 2x + 1)$
 B $(x - y)(c + f)$
 C $20(2m - n)(2m + n)$
 D $x(x - 6)$
 E $(x + y)(x + 7)$
 F $5y(x + 3z)$
 G $(m - n)(m + n + 5)$
 H $5(2a + 5b)$
 I $6y(2 - y)$
 J $(1 - 9a)(1 + 9a)$
 K $(a - 8)(a - 3)$
 L $(b - 1)(b^2 + b + 1)$
 M $(3a - 2b)(x + y)$
 N $(3x - 7y)(3x + 7y)$
 O $(a + 5)(a^2 - 5a + 25)$
 P $3x(x + 4y)$
 Q $(x - 7)(x + 2)$
 R $7a(3a + b - 5)$
 S $(3 - y)(9 + 3y + y^2)$
 T $(x - 5)(x^2 + 4)$
 U $(a - b)(x - 3)$
 V $y(ay + 4a - 3)$
 W $(6a - 5)(7a + 3)$
 X $(x - 7y)(x + y)$
 Y $(6 + m)(2x - y)$
 Z $(x - 1)(x^2 - 9) = (x - 1)(x - 3)(x + 3)$
 AA $2(x - 6)(7x + 2)$
 BB $6(a + 1)(a^2 - a + 1)$
 CC $(x - 7)(x + 1)$
 DD $9(y + 2)(y^2 - 2y + 4)$
 EE $(8x + 1)(1 - 9x)$
 FF $(x - 1)(x + 1)(x^2 + x + 1)(x^2 - x + 1)$ or $(x - 1)(x + 1)(x^4 + x^2 + 1)$
 GG $(ab - 1)(ab + 1)$

HH $(5x - 3)(2x + 1)$

II $(x^2 + 9)(x - 3)(x + 3)$

JJ $2(a + 7)(5a + 1)$

5-2, 5-3, 5-4, 5-5, 5-6 No answers required

5-7 Graph classification — cardmatch

Straight line: 43; 1, 6, 14, 15, 19, 23, 27, 35

Parabola: 45; 2, 7, 9, 17, 33, 36

Hyperbola: 44; 3, 8, 12, 20, 24, 29, 32

Circle or semicircle: 42; 4, 11, 18, 22, 26, 28

Cubic: 41; 5, 13, 34, 39

Exponential: 47; 10, 16, 30

Logarithmic: 48; 21, 31, 38

Absolute value: 46; 25, 37, 40

5-8 Matching graphs — cardmatch

1 C	2 E	3 L	4 A
5 R	6 F	7 B	8 N
9 M	10 D	11 H	12 J
13 U	14 K	15 V	16 G
17 T	18 P	19 Q	20 S

5-9 Derivatives — cardmatch

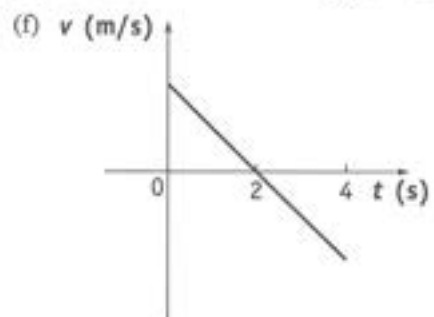
1 C	2 K	3 A	4 J
5 B	6 D	7 G	8 I
9 F	10 E	11 H	12 L

5-10 Discriminant — cardmatch

A 1, 5, 14, 15	B 17, 13, 4, 3, 20
C 16, 13, 7, 2	D 12, 19, 11, 21
E 18, 6, 23, 8, 22	F 9, 5, 14, 10

5-11 Investigating motion

- 1 (a) $t = 0$ s and $t = 4$ s
 (b) 5 m (c) 10 m
 (d) 2.5 m/s (e) $t = 2$ s



(g) Teacher to check

- 2 (a) 1 m (b) $t = 7$ s
(c) 5 m (d) 12 m
(e) $t = 3$ s and $t = 7$ s
(f) Teacher to check

- 3 (a) $t = 0$ s, $t = 4.75$ s and $t = 7$ s
(b) $t = 2$ s and $t = 6$ s
(c) -2 m/s
(d) between $t = 2$ s and $t = 6$ s
(e) C

- 4 (a) $t = 5$ s
(b) between $t = 0$ s and $t = 4$ s
(c) -2 m/s² (d) 13 m
(e) 5 m (9 m - 4 m) to the right of the starting position and moving in a negative direction.

- 5 (a) No. It will always be travelling in a positive direction.
(b) The velocity increases.
(c) The initial position is to the left of the origin.

- 6 (a) C (b) $t = 3$ seconds
(c) Negative. It started off moving to the right but some time after 3 s it changed direction.

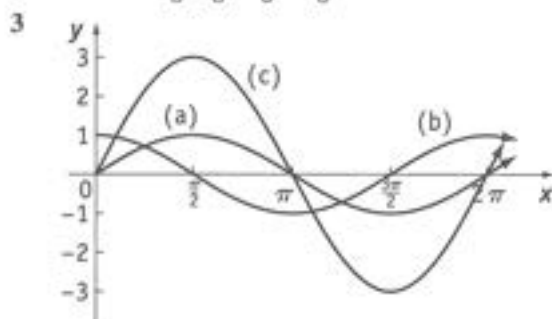
- (b) $x = 12$ and $y = 36$ (which is half of 72 m)

6 Teacher to check

- 7 (a) $a < -1, a > 4$ (b) $-1 < b < 5$
(c) $-\frac{7}{3} \leq x < 7$ (d) $x = -3.5$

5-16 Study sheet 2

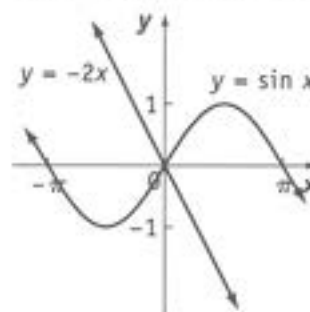
- 1 (a) 0.841 (b) 0.878
2 (a) $\theta = \frac{\pi}{3}, \frac{5\pi}{3}$ (b) $\theta = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$
(c) $\theta = \frac{\pi}{8}, \frac{3\pi}{8}, \frac{9\pi}{8}, \frac{11\pi}{8}$



- 4 $(y - 3)^2 = -8(x - 2)$
5 $P = 2a + \pi a + 2a\theta, A = \frac{1}{2}\pi a^2 + 2a^2\theta$
6 (a) $A = \frac{14\,175\pi}{8} \text{ cm}^2$
(b) $P = 90 + \frac{315\pi}{4} \text{ cm}$
7 (a) 100 hours
(b) \$17 000
(c) $C = \frac{160\,000}{V} + 10V^2$
(d) 20 km/h, \$12 000
(e) \$12 560

5-17 Study sheet 3

- 1 (a) $-2 \sin 2x$ (b) $3 \sin^2 x \cos x$
(c) $\frac{4x \cos 4x - \sin 4x}{x^2}$
2 (a) $x - y = 4$ (b) $x^2 = -12y + 36$
(c) $x^2 + y^2 - 12y + 32 = 0$
3 (a) Only one point of intersection:



5-12 No answers required

5-13 A series of questions

- 1 \$187 355
2 10 years
3 5.556 mg
4 (a) to (c) Teacher to check
(d) $A_{100} = \$26\,314$
(e) interest = \$16 314, total repaid = \$90 000
percentage interest = 18.1%

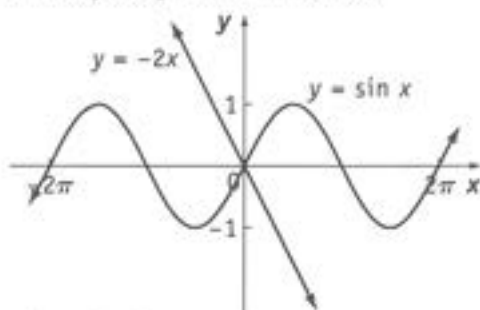
5-14 Parabola language — cardmatch

- | | | |
|-----|-----|-----|
| 1 C | 2 A | 3 F |
| 4 B | 5 G | 6 D |
| 7 E | 8 H | |

5-15 Study sheet 1

- 1 (a) $(x - 1)^2 = 12(y - 4)$
(b) $y^2 = 8(x - 1)$
2 4.5 units²
3 $1\frac{2}{3}$ units²
4 $4y = 13x - 25$
5 (a) Teacher to check

- (b) Still only one point of intersection:



4 $x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$

5 0.64

6 0.5002

7 $\theta = 0.77^\circ = 0.8^\circ$

8 Teacher to check

9 (a) $x \leq -3, x \geq 1$

(b) $m \leq -3, m \geq 1$

5-18 Study sheet 4

1 (a) $\sin 2x + 2x \cos 2x$

(b) $\frac{3 \cos x + 3x \sin x}{\cos^2 x}$

(c) $2 \tan x \sec^2 x$

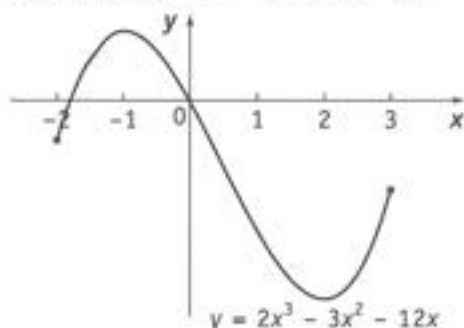
2 $y - 2 = 2\sqrt{3}(x - \frac{\pi}{6})$

3 $(\frac{7\pi}{6}, -\frac{\sqrt{3}}{2})$ and $(\frac{11\pi}{6}, \frac{\sqrt{3}}{2})$

4 1.2

5 (a) (2, -20) is the minimum turning point;
(-1, 7) is the maximum turning point.

(b) Endpoints are (-2, -4) and (3, -9):



(c) $-20 \leq y \leq 7$

6 Teacher to check ($R = \frac{240}{\pi + 4}$)

7 (a) $\frac{-2}{x^3}$

(b) Teacher to check

5-19 Study sheet 5

1 (a) $2(4x + 2)^{\frac{1}{2}}$

(b) $\cos^2 x - \sin^2 x$

(c) $\cos 4x - 4x \sin 4x$

2 (a) $\frac{1}{9}(3x - 1)^3 + C$

(b) $\frac{1}{5}x^5 - \frac{2}{3}x^3 + x + C$

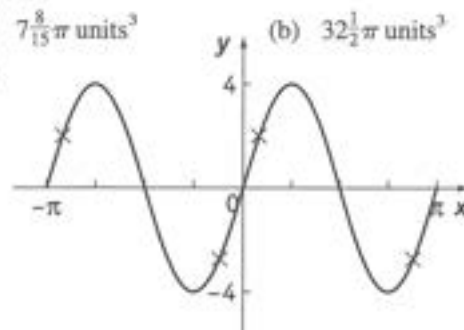
(c) $-\frac{10}{x} + C$

3 (a) $\operatorname{cosec} 0 = \infty$ and $\operatorname{cosec} \pi = \infty$.
The area is not bounded.

(b) 1.9

4 (a) $7\frac{8}{15}\pi \text{ units}^3$

5 (a)



(b) period = π , amplitude = 4

(c) See graph (x)

(d) In $-\pi \leq x \leq \pi$, $x = -2.42, -0.72, 0.72, 2.42$

6 (a) $A(\frac{\pi}{4}, 0.7), B(\frac{5\pi}{4}, -0.7)$

(b) Area = $(\frac{4}{\sqrt{2}}) + (\frac{2}{\sqrt{2}} + 1)$
 $= 3\sqrt{2} + 1 \text{ units}^2$

5-20 Study sheet 6

1 Teacher to check

2 (a) Teacher to check

(b) $BE = 22.5 \text{ cm}$

3 (a) 3^{9n} (b) $2^{3n+2} \times 3^2$

4 $V(2, -\frac{1}{3}), F(2, \frac{2}{3}), \text{directrix } y = -3\frac{1}{3}$

5 discriminant = $9b^2 + 4c^2$, which is always > 0

6 $k = 0$, roots -1 and 0; $k = 7$, roots 6 and 7

7 0.3

8 (a) 3

(b) $\frac{1}{2}$

(c) 6

(d) 1.5 (e) 8

5-21 Study sheet 7

1 Teacher to check

2 (a) Teacher to check

(b) $AC = 26\frac{2}{3} \text{ cm}$

3 $m = 1.5$, roots $\frac{1}{2}$ and 1; $m = 9$, roots -2, -4

4 $c = a$

5 (a) Teacher to check. Note that focus is (0, 2).

(b) At P, $9y = -3x - 2$; at Q, $y = 3x - 18$

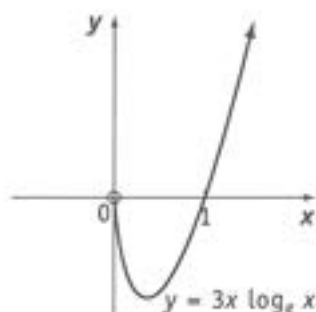
(c) (i) $3 \times -\frac{1}{3} = -1$

(ii) Teacher to check. Note that directrix is $y = -2$.

- 6 (a) Teacher to check (b) -3

5-22 Study sheet 8

- 1 3^{6n-4}
 2 (a) $x = 16$ (b) $x = 3$ (c) $x = \frac{5}{4}$
 3 2.89
 4 (a) minimum turning point at $\left(\frac{1}{e}, -\frac{3}{e}\right)$
 (b) $x > 0$
 (c)

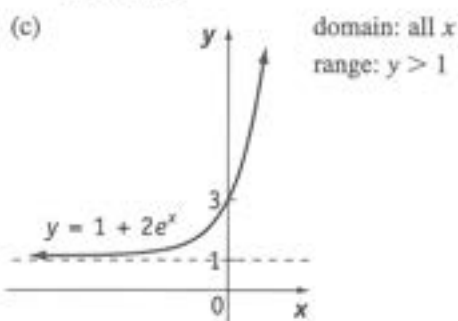
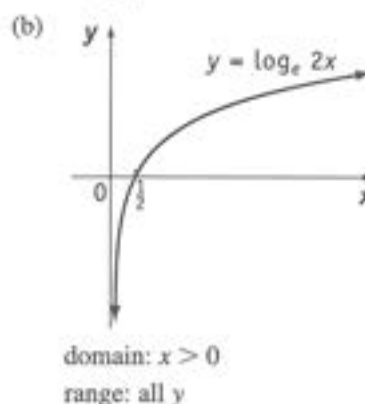
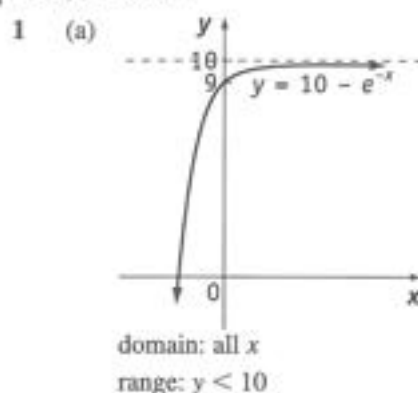


- (d) $y \geq -\frac{3}{e}$
 5 $\frac{5x-11}{(x-2)(x+2)}$
 6 (a) The diagonals of a parallelogram bisect each other.
 (b) Teacher to check
 7 Teacher to check
 8 $t = 3.35$
 9 $x < 0, x > 5$
 10 $\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$

- (b) $y = \sin 2x$: period $= \pi$, amplitude $= 1$
 $y = 2 \cos x$: period $= 2\pi$, amplitude $= 2$

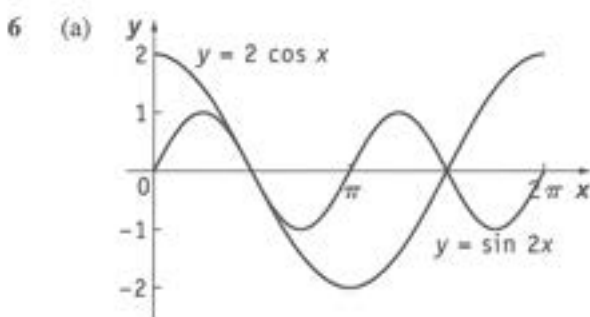
- (c) 4 units²
 7 (a) $\theta = \frac{\pi}{12}, \frac{11\pi}{12}, \frac{13\pi}{12}, \frac{23\pi}{12}$
 (b) $\theta = \frac{\pi}{3}, \frac{4\pi}{3}$
 8 0.84
 9 (a) 3^{2x} (b) 3^{x+1}
 (c) 3^{2x} (d) 9^{8x}

5-24 Study sheet 10

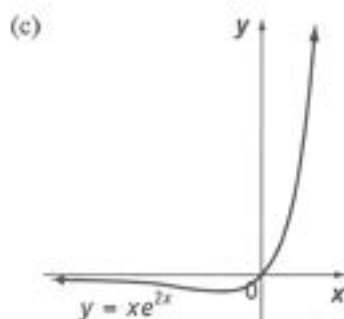


5-23 Study sheet 9

- 1 (a) In an isosceles trapezium, base angles are equal.
 (b) and (c) Teacher to check
 2 (a) Teacher to check
 (b) $AC = 38.4$ cm, $BC = 18.4$ cm
 3 (a) $y = x - 1$
 (b) 0.5 units²
 4 $x = 3$
 Note that $x \neq -2$ as $\log(-2)$ does not exist.
 5 2.5



- 2 Point $\left(\frac{1}{3}, \log_e \frac{2}{3}\right)$
 3 Teacher to check
 4 (a) Teacher to check (b) $e^2 - 1$ units²
 5 (a) Teacher to check
 (b) $\left(-\frac{1}{2}, -\frac{1}{2}e^{-1}\right)$ is a minimum turning point.



(d) $y \geq -\frac{1}{2}e^{-1}$

6 $K = 3.5$

7 $\pi(\frac{1}{2}e^6 + 2e^3 - \frac{1}{2}e^2 - 2e + 2)$ units³

8 $x = \frac{17}{13}$

9 $\frac{1}{3}\sin^3 x + C$

10 (a) Teacher to check (b) 1

5-25 Study sheet 11

1 (a) $V(1, -2), F(1, 1), y = -5$

(b) $V(-3, 4), F(-3, -0.5), y = 8.5$

(c) $V(1, 0), F(3, 0), x = -1$

(d) $V(1, 2), F(\frac{1}{2}, 2), x = \frac{1}{2}$

(e) $V(0, -5), F(0, -3), y = -7$

(f) $V(0, -12), F(0, -11\frac{3}{4}), y = -12\frac{1}{4}$

2 (a) $F(5, 3), y = -9$

(b) $F(0, -2), x = 2$

(c) $F(-3, 0), y = -3$

3 (a) $(x - 2)^2 = 16(y - 5)$

(b) $(x - 3)^2 = -24(y - 6)$

(c) $(y - 5)^2 = 4(x - 2)$

(d) $(y - 5)^2 = -12(x - 3)$

(e) $x^2 = -8(y - 6)$

4 $2y = x - 1$

5 $2y = x - 36$

6 $x^2 = 20(y - 5)$

7 (a) Teacher to check

(b) (4, 5)

8 (a) $V(0, 3)$ (b) $y = 5$

(c) $(-4, 5), (4, 5)$ (d) $10\frac{2}{3}$ units²

9 (a) $(x - 2)^2 = 2(y + 4)$

(b) (2, -3.5)

5-26 Study sheet 12

1 (a) $x^3 - 5x + C$ (b) $x^4 + 3x^2 + 5x + C$

(c) $\frac{(3x + 4)^6}{18} + C$ (d) $\frac{(6x + 7)^{1.5}}{9} + C$

2 99

3 $49\frac{1}{3}$ units²

4 57π units³

5 (a)

x	0	0.5	1	1.5	2
y	2	1.936	1.732	1.323	0

(b) 3.08

(c) Calculate the area of a quarter of a circle with radius 2.

6 (a) $A = -3B^{-2} + 13$

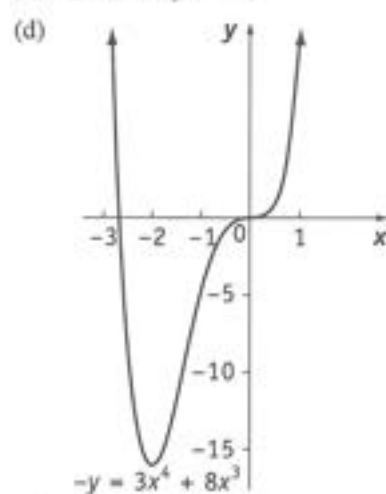
(b) $f(x) = 2x^2 - x - 3$

7 (a) $1\frac{1}{3}$ units² (b) 4 units²

8 (a) $(-2, -16), (0, 0)$

(b) $(-2, -16)$ is a minimum turning point and $(0, 0)$ is a horizontal point of inflection.

(c) $(0, 0), (-\frac{4}{3}, -9.5)$



9 (a)

x	0	40	80	120	160
y	110	150	100	75	125

(b) 17 700 m²

10 (a) $-\frac{1}{5}$ (b) $-6 < m < 6$

11 (a) 1 (b) $\frac{4}{3}$ (not 0)

12 48 (not 56)

13 (a) $V(4, -1), F(4, -6)$ (b) $y = 4$

(c) $x = 4$

14 (a) $V(1.5, 4), F(-1, 4)$ (b) $x = 4$

- 15 (a) $\sqrt{20}$ units (b) $-\frac{1}{2}$
 (c) 2 (d) Teacher to check
 (e) $F(-10, 0)$ (f) Teacher to check

5-27 Study sheet 13

- 1 (a) $2 \cos 2x$ (b) $\frac{1}{2} \sec^2 \frac{x}{2}$
 (c) $2x + \pi \sin \pi x$ (d) $3 - 2e^{-2x}$
 (e) $3 \log_e (x^2 - 7) + \frac{6x^2}{x^2 - 7}$
 (f) $-4e^{-x}$ (g) $-30(2 - 3x)^9$
 (h) $\frac{2x \sin x - x^2 \cos x}{\sin^3 x}$ (i) π

- 2 (a) $x^2 + \frac{1}{3}e^{3x} + C$
 (b) $-\frac{2\pi}{3} \cos 3x + C$
 (c) $3 \tan \frac{x}{3} + C$
 (d) $\frac{3}{2} \log_e (x^2 + 5) + C$
 (e) $\frac{2}{3}x^2 \sqrt{x} + C$
 (f) $\log_e x - e^{-x} + C$

3 2.07

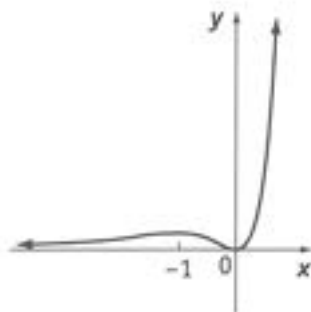
- 4 (a) Teacher to check
 (b) $6 + 3 \log_e 4$ or $6(1 + \log_e 2)$

5 Teacher to check

- 6 (b) $(0, -1)$ and $(-\frac{1}{2}, 0)$ (c) $\frac{1}{4} \text{ unit}^2$

- 7 (a) $A = \left(\frac{3\pi}{4}, \frac{1}{\sqrt{2}}\right)$, $B = \left(-\frac{\pi}{4}, -\frac{1}{\sqrt{2}}\right)$
 (b) $2\sqrt{2} \text{ units}^2$

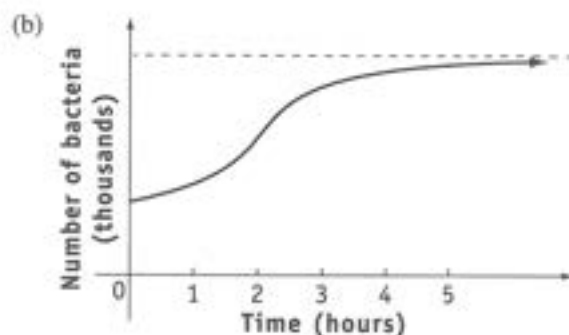
- 8 (a) $(0, 0)$ (b) $4xe^{2x}(1 + x)$
 (c) $(-1, 2e^{-2})$ is a maximum turning point and
 $(0, 0)$ is a minimum turning point.
 (d) As $x \rightarrow \infty$, $y \rightarrow \infty$, and as $x \rightarrow -\infty$, $y \rightarrow 0$.
 (e)



- 9 (a) Teacher to check
 (b) $4 \log_e 4 - 3$ (or 2.55) units^2

- 10 (a) Teacher to check
 (b) $y = 4e^{2x}$

- 11 (a) 6670 bacteria



5-28 Study sheet 14

- 1 $x = -\frac{37}{4}$

- 2 (a) $x \leq -4$, $x \geq 5$

- (b) $0 < x < 1$

- (c) $-5 < x < 4$

- 3 (a) $\frac{99}{64}$ (b) $-7\frac{1}{2}$ (c) $-\frac{3}{8}$

- 4 $K = \frac{25}{8}$

- 5 \$1135.21

- 6 (a) Teacher to check

- (b) $\frac{\pi}{4}(e^2 - 1) \text{ units}^3$

- 7 (a) 450 m^3 (b) 14 days (c) $13.6 \text{ m}^3/\text{day}$

- 8 (a) $y = \frac{1}{2}, -3$

- (b) $x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}, \frac{17\pi}{6}$

- (c) $t = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}, \frac{17\pi}{6}$ seconds
 or = 0.5, 2.6, 6.8, 8.9 seconds

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